



NATIONAL BOARD FOR TECHNICAL EDUCATION

**HIGHER NATIONAL DIPLOMA
(HND)**

NETWORKING AND CLOUD COMPUTING

CURRICULUM AND COURSE SPECIFICATION

AUGUST, 2023

PLOT B, BIDA ROAD, P.M.B. 2239, KADUNA – NIGERIA

www.nbte.gov.ng

Foreword

The Higher National Diploma (HND) Networking and Cloud Computing curriculum is designed to be used by training institutions to produce manpower to boost the practice of Networking and Cloud Computing in Nigeria.

There is therefore the need to produce professional practitioners with good ethics and prospects for career progression, through the acquisition of desirable knowledge and skills. This necessitated the production of this national curriculum.

It is my belief that this curriculum and course specifications, which is the minimum required to produce technologists with sound knowledge and skills in Networking and Cloud Computing, if properly implemented with the required resources along with qualified candidates being admitted into the programme, will lead to the production of competent and skilled technologists.

I sincerely appreciate all those who made the development of this curriculum possible. The invaluable contributions of all the members of the committee and resource persons during the pre-critique and national critique workshops are appreciated.

I hope that the curriculum would be properly implemented so as to produce the required Work Force of our dream.

Prof. Idris M. Bugaje
Executive Secretary
NBTE, Kaduna

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GENERAL INFORMATION

1.0 TITLE OF THE PROGRAMME:

Higher National Diploma Networking and Cloud Computing

2.0 GOAL AND OBJECTIVES:

2.1 GOAL: At the end of the programme the students should be able to show competency in Networking and Cloud Computing

2.2 OBJECTIVES OF THE PROGRAMME:

A diplomat of HND Networking and Cloud Computing should be able to:

- i. Design, build, secure, and manage networks
- ii. Deploy Cloud Computing services
- iii. Use Operating Systems in Network Environment
- iv. Design and deploy Internet of Things (IOT) solutions
- v. Give professional advice to clients on system specifications
- vi. Use Python Programming to solve real-life problems
- vii. Apply Cloud solutions to solve organizational problems across various industries
- viii. Obtain industry-based certification
- ix. Conduct independent research
- x. Deploy solutions in Virtualized environment
- xi. Apply Ethical & Professional Practice in Networking and Cloud Computing
- xii. Set up and manage a business.

3.0 ENTRY REQUIREMENTS

The general entry requirements for the HND Networking and Cloud Computing Programme are:

- (a) In addition to the entry requirement for National Diploma.
- (b) ND Computer Science or National Skills Qualification (NSQ) level 3-5 in Networking and Cloud Computing
- (c) One (1) year relevant industrial training
- (d) NID Networking and System Security
- (e) ND Diplomats with a pass (CGPA of 2.0-2.49) in the ND examination with two or more years of cognate experience in the specific field may be considered.

4.0 CURRICULUM

5.0 STRUCTURE OF PROGRAMME

This is a two-year Programme i.e. four semesters of classroom, laboratory, field and workshop activities in the institution. Each semester shall be of 17 weeks duration made up as follows:

15 Contact weeks of teaching, i.e. recitation, practical exercises, quiz, tests, etc and 2 weeks for examination and registration.

5.0 EVALUATION SCHEME

The HND Networking and Cloud Computing Examination must be externally moderated. In grading the students, theory shall constitute 40% while practical and project are 60% respectively.

6.0 ACCREDITATION

Each Programme offered at the HND level shall be accredited by the NBTE before the Diplomates can be awarded the Higher National Diploma certificates. Details about the process of accrediting a Programme for the award of the HND are available from the office of the Executive Secretary, National Board for Technical Education, Plot B, Bida Road, P.M.B. 2239, Kaduna, Nigeria.

7.0 CONDITIONS FOR THE AWARD OF HND NETWORKING AND CLOUD COMPUTING

Institutions offering this Programme will award the HND certificate to candidates who successfully completed the Programme after passing prescribed course work, examinations, Project and Industry-based certification as approved by NBTE. Such candidates should have completed a minimum of 85 semester credit units.

7.1 GRADING OF COURSES: Courses shall be graded as follows:

MARKED RANGE	LETTER GRADE	WEIGHTING
75% and above	A	4.00
70% – 74%	AB	3.50
65% – 69%	B	3.25
60% – 64%	BC	3.00
55% – 59%	C	2.75
50% – 54%	CD	2.50
45% – 49%	D	2.25
40% – 44%	E	2.00

7.2 CLASSIFICATION OF DIPLOMAS: Higher National Diploma Certificates shall be awarded based on the following classifications:

Distinction	-	CGPA 3.50-4.00
Upper Credit	-	CGPA 3.00-3.49
Lower Credit	-	CGPA 2.50-2.99
Pass	-	CGPA 2.00-2.49

8.0 QUALIFICATION OF THE TEACHERS:

8.1 Holders of B.Sc/HND or a higher degree qualifications in Computer Science related disciplines

8.2 In addition, teachers of this programme should have been trained and certified by recognized professional and industrial bodies in Networking and Cloud Computing.

9.0 GUIDANCE NOTES FOR TEACHERS OF THE PROGRAMME

9.1 The new curriculum is drawn in unit courses. This is in keeping with the provisions of the National Policy on Education which stresses the need to introduce the semester credit units which will enable a student who so wish to transfer the units already completed in an institution of similar standard from which he is transferring.

9.2 In designing the units, the principle of the modular system by product has been adopted; thus, making each of the professional modules, when completed provides the student with technician operative skills, which can be used for employment purpose.

As the success of the credit unit system depends on the articulation of Programme between the institutions and industry, the curriculum content has been written in behavioral objectives, so that it is clear to all the expected performances of the student who successfully completed some of the courses or the diplomates of the Programme. There is a slight departure in the presentation of the performance-based curriculum which requires the conditions under which the performance is expected to be carried out and the criteria for the acceptable levels of performance. It is a deliberate attempt to further involve the staff of the department teaching the Programme to write their own curriculum stating the conditions existing in the institution under which the performance can take place and to follow that with the criteria for defining an acceptable level of performance. Departmental submission on the final curriculum may be vetted by the Academic Board of the institution.

9.3 Our aim is to continue to see to it that a solid internal evaluation system exists in each institution for ensuring minimum standard and quality of education in the programmes offered throughout the Technical and Vocational Education (TVE) system.

9.4 The teaching of the theory and practical work should as much as possible be integrated. Practical exercises, especially those in professional courses and laboratory work should not be taught in isolation from the theory. For most courses, there should be a balance of theory to practice in the ratio of 40:60

10. MANDATORY SKILLS QUALIFICATION:

CURRICULUM TABLE

YEAR ONE SEMESTER ONE

S/N	COURSE CODE	COURSE TITLE	L	P	CU	CH
1	GNS 301	Use of English III	2	0	2	2
2	NCC 311	Networking Essentials	2	2	3	4
3	NCC 312	Virtualization and Cloud Migration	2	2	3	4
4	NCC 313	Operating System for Network Environment	1	2	3	3
5	NCC 314	Wireless Communication Systems	2	2	3	4
6	NCC 315	Computer Architecture	2	1	3	3
7	NCC 316	Python Programming Basics	1	2	2	3
Total			12	11	19	23

YEAR ONE, SEMESTER TWO

S/N	COURSE CODE	COURSE TITLE	L	P	CU	CH
1	GNS 302	Communication in English III	2	0	2	2
2	ENT 326	Practice of Entrepreneurship 1	2	2	2	4
3	AIT 313	Artificial Intelligence	2	2	3	4
4	CYS 322	Mobile Wireless Security	2	2	4	4
5	NCC 321	Routing and Switching 1	1	3	4	4
6	NCC 322	Cloud Computing 1	1	2	3	3
7	NCC 323	Advance Statistics for Computing	1	2	3	3
8	NCC 324	Mandatory Skill Qualification 1	0	4	4	4
9	NCC 325	Advanced Wireless Networks	2	2	3	4
Total			13	19	28	32

YEAR TWO SEMESTER ONE

S/N	COURSE CODE	COURSE TITLE	L	P	CU	CH
1	GNS 401	Communication in English IV	2	0	2	2
2	ENT 416	Practice of Entrepreneurship II	2	2	4	4
3	AIT 321	Machine Learning	2	2	3	4
4	CYS 412	Data Protection and Privacy	2	1	3	3
5	NCC 411	Research Methodology	2	0	2	2
6	NCC 412	Project Management in Networking & Cloud Computing	1	1	2	2
7	NCC 413	Routing and Switching II	1	3	4	4
8	NCC 414	Networking Troubleshooting & Monitoring	1	2	3	3
9	NCC 415	Mandatory Skills Qualifications II	0	4	4	4
Total			13	15	27	28

YEAR TWO, SEMESTER TWO

S/N	COURSE CODE	COURSE TITLE	L	P	CU	CH
1.	NCC 421	Cloud Computing II	1	2	3	3
2.	NCC 422	Enterprise Networking, Security & Automation	1	3	4	4
3.	NCC 423	Ethical & Professional Practice in Networking and Cloud Computing	2	0	2	2
4.	NCC 424	Internet of Things	1	2	3	3
5.	NCC 425	Seminar	2	0	2	2
6.	NCC 426	Project	0	6	6	6
Total			7	13	20	20

KEY: L= Lecture hours, T= Tutorial hours, P= Practical hours, CU= Credit Units, CH= Contact Hour

YEAR ONE SEMESTER ONE COURSES

NETWORKING ESSENTIALS

PROGRAMME: HIGHER NATIONAL DIPLOMA (HND) NETWORKING AND CLOUD COMPUTING		
COURSE: NETWORKING ESSENTIALS	COURSE CODE: NCC 311	Contact Hours: 3 hours /week
	CREDIT UNITS: 3	Theoretical: 2 hours /week
Year: 1 Semester: 1	Pre-requisite:	Practical: 1 hour/week
GOAL: This course is designed to equip students with networking skills.		
GENERAL OBJECTIVES: On completion of the course, the student should be able to: 1.0 Understand Networks Today 2.0 Understand Protocols and Models 3.0 Know Open System Interconnect (OSI) model 4.0 Understand Ethernet Switching 5.0 Understand Number Systems and IP addressing 6.0 Understand Basic Router and Switch Configuration 7.0 Understand Network Security Fundamentals 8.0 Know how to build Small Networks		

PROGRAMME: HIGHER NATIONAL DIPLOMA (HND) NETWORKING AND CLOUD COMPUTING						
COURSE: NETWORKING ESSENTIALS		COURSE CODE: NCC 311			Contact Hours: 3hours/Week	
		CREDIT UNIT: 3			Theoretical: 2 Hours/Week	
Year: I Semester: I		Pre-requisite:			Practical: 1 Hour/Week	
GOAL: This course is designed to equip students with networking skills.						
GENERAL OBJECTIVE: 1.0 Understand networks today						
THEORETICAL CONTENT				PRACTICAL CONTENT		
Week	Specific Learning Outcome	Teacher’s Activities	Resources	Specific Learning Outcome	Teacher’s Activities	Evaluation
1-2	1.1 Explain how networks affect our daily lives.	Explain how networks affect our daily lives.	Multi-media Projector, Lecture notes, Tutorials, Pictorials Packet tracer or any other suitable simulator	Identify network devices	Demonstrate Network Representation using simulation software.	Outline network representations and how they are used in network topologies List the characteristics of a good network Describe types of network topologies
	1.2 Outline how host and network devices are used	Explain how host and network devices are used		Design LAN/WAN using the simulator	Demonstrate how to design LAN/WAN using the simulator	
	1.3 Outline network representations and how they are used in network topologies.	Explain network representations and how they are used in network topologies.		Design types of network topologies using the simulator	Demonstrate how to design network topologies using the simulator	
	1.4 List the characteristics of common types of networks.	Compare the characteristics of common types of networks.				
	1.5 Outline how LANs and WANs	Explain how LANs and WANs interconnect to the				

	interconnect to the internet. 1.6 Explain the four basic requirements of a reliable network. 1.7 Outline how trends such as BYOD, online collaboration, video, and cloud computing are changing the way we interact 1.8 Explain some basic security threats and solutions for all networks 1.9 Outline employment opportunities in the networking field	internet. - Describe the four basic requirements of a reliable network. - Explain how trends such as BYOD, online collaboration, video, and cloud computing are changing the way we interact - Identify some basic security threats and solution for all networks - Explain employment opportunities in the networking field				
• GENERAL OBJECTIVE 2.0: Understand protocols and models						
3-4	2.1 Outline the types of rules that are necessary to successfully communicate. 2.2 Narrate why	- Describe the types of rules that are necessary to successfully communicate. - Explain why	Multi-media Projector, Lecture notes, Tutorials, Pictorials Wireshark packet tracer or	Investigate the TCP/IP and OSI Models in Action using packet tracer Install Wireshark to view network traffic.	Guide students to; - Investigate the TCP/IP and OSI Models in Action using packet tracer - Install Wireshark to view Network	Explain why protocols are necessary in network communication Explain the

	protocols are necessary for network communication 2.3 Outline the purpose of adhering to a protocol suite. 2.4 Outline the role of standards organizations in establishing protocols for network interoperability. 2.5 Outline how the TCP/IP model and the OSI model are used to facilitate standardization in the communication process. 2.6 Outline how data encapsulation allows data to be transported across the network. 2.7 Outline how	protocols are necessary in network communication • Explain the purpose of adhering to a protocol suite. • Explain the role of standards organizations in establishing protocols for network interoperability. • Explain how the TCP/IP model and the OSI model are used to facilitate standardization in the communication process. • Explain how data encapsulation allows data to be transported across the network. • Explain how local	other network simulators	Apply ping and traceroute utilities to test network connectivity Use ping and traceroute utilities to test network connectivity	Traffic • Apply ping and trace route utilities to test network connectivity • Use ping and trace route utilities to test network connectivity	purpose of adhering to a protocol suite.
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	local hosts access local resources on a network. 2.8 Contrast the roles of the MAC address and the IP address. 2.9 Explain the purpose of ARP. 2.10 Explain the operation of IPv6 neighbour discovery. 2.11 Outline how ICMP is used to test network connectivity. 2.12 Explain the techniques of ensuring successful communication between devices	hosts access local resources on a network. • Compare the roles of the MAC address and the IP address. • Describe the purpose of ARP. • Describe the operation of IPv6 neighbour discovery. • Explain how ICMP is used to test network connectivity. • Use ping and traceroute utilities to test network connectivity				
• GENERAL OBJECTIVE: 3.0 Know Open System Interconnect (OSI) model						
5-7	3.1 Explain the purpose and functions of the physical layer in	• Elaborate the purpose and functions of the physical layer in	Multi-media Projector, Lecture notes, Tutorials,	• Make crossover and straight-through cable • Connect wired	Guide students to: • Make crossover and straight-through cable	List the characteristics of media access control methods

	the network. 3.2 Explain the characteristics of the physical layer. 3.3 Explain the basic characteristics of copper cabling. 3.4 Explain how UTP cable is used in Ethernet networks. 3.5 Explain fiber optic cabling and its main advantages over other media. 3.6 Explain Network devices using wired and wireless media. 3.7 List the purpose and function of the data link layer in preparing communication for transmission on specific media.	the network. • Elaborate the characteristics of the physical layer. • Explain the basic characteristics of copper cabling. • Describe how UTP cable is used in Ethernet networks. • Explain fiber optic cabling and its main advantages over other media. • Explain Network devices using wired and wireless media. • List the purpose and function of the data link layer in preparing communication for transmission on specific media.	Pictorials Wireshark packetacer or other network simulators networking tool kit networking cable	and wireless LAN. • View Wired and Wireless NIC Information using Packet Tracer. • Observe TCP and UDP Communications • Observe DNS Resolution	• Connect a Wired and Wireless LAN • View Wired and Wireless NIC Information Packet Tracer - TCP and UDP Communications • Observe DNS Resolution	on WAN and LAN topologies
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	3.8 List the characteristics of media access control methods on WAN and LAN topologies. 3.9 Explain the characteristics and functions of the data link frame. 3.10 Explain how the network layer uses IP protocols for reliable communications 3.11 Explain the role of the major header fields in the IPv4 packet. 3.12 Explain the role of the major header fields in the IPv6 packet. 3.13 Explain how network devices use routing tables to direct packets to a destination network.	• List the characteristics of media access control methods on WAN and LAN topologies. • Explain the characteristics and functions of the data link frame. • Explain how the network layer uses IP protocols for reliable communications • Explain the role of the major header fields in the IPv4 packet. • Explain the role of the major header fields in the IPv6 packet. • Explain how network devices use routing tables to direct packets to a destination network.				
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	3.14 Explain the functions of fields in the routing table 3.15 Explain the purpose of the transport layer in managing the transportation of data in end-to-end communication. 3.16 List the characteristics of TCP. 3.17 List characteristics of UDP. 3.18 Describe how TCP and UDP use port numbers. 3.19 Explain how TCP session establishment and termination processes facilitate reliable communication. 3.20 Explain how	• Explain the function of fields in the routing table of a router. • Explain the purpose of the transport layer in managing the transportation of data in end-to-end communication. • List the characteristics of TCP. • List characteristics of UDP. • Describe how TCP and UDP use port numbers. • Explain how TCP session establishment and termination processes facilitate reliable communication. • Explain how TCP				
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	TCP protocol data units are transmitted and acknowledged to guarantee delivery. 3.21 Explain the operations of transport layer protocols in supporting end-to-end communication. 3.22 Explain how the functions of the application layer, presentation layer, and session layer work together to provide network services to end-user applications. 3.23 Explain how end-user applications operate in a peer-to-peer network. 3.24 Explain how web and email protocols	protocol data units are transmitted and acknowledged to guarantee delivery. • Explain the operations of transport layer protocols in supporting end-to-end communication. • Explain how the functions of the application layer, presentation layer, and session layer work together to provide network services to end-user applications. • Explain how end-user applications operate in a peer-to-peer network. • Explain how web and email protocols operate.				
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	operate. 3.25 Explain how DNS and DHCP operate. 3.26 Explain how file transfer protocols operate.	• Explain how DNS and DHCP operate. • Explain how file transfer protocols operate.				
GENERAL OBJECTIVE: 4.0 Understand Ethernet switching						
8-9	4.1 Describe how the Ethernet sublayers are related to the frame fields. 4.2 Describe the Ethernet MAC address. 4.3 Describe how a switch builds its MAC address table and forwards frames. 4.4 Describe switch forwarding methods and port settings available on Layer 2 switch ports.	• Explain how the Ethernet sublayers are related to the frame fields. • Describe the Ethernet MAC address. • Explain how a switch builds its MAC address table and forwards frames. • Describe switch forwarding methods and port settings available on Layer 2 switch ports.	Multi-media Projector, Lecture notes, Tutorials, Pictorials Multi-media Projector, Lecture notes, Tutorials, Pictorials Computer Routers Switches	• Use Wireshark to Examine Ethernet Frames • View Network *Device MAC Addresses • View the Switch MAC Address Table	Guide students to; • Use Wireshark to Examine Ethernet Frames • View Network Device MAC Addresses • View the Switch MAC Address Table	Describe how a switch builds its MAC address table and forwards frames

GENERAL OBJECTIVE: 5.0 Understand Number systems and IP addressing						
10	5.1 Explain how to convert numbers between decimal and binary systems. 5.2 Explain how to convert numbers between decimal and hexadecimal systems. 5.3 Explain the structure of an IPv4 address including the network portion, the host portion, and the subnet mask. 5.4 Analyse the characteristics and uses of the unicast, broadcast and multicast IPv4 addresses. 5.5 Describe public, private, and reserved IPv4 addresses. 5.6 Describe how subnetting segments a network to	• Explain how to convert between decimal and binary systems. • Explain how to convert numbers between decimal and hexadecimal systems. • Describe the structure of an IPv4 address including the network portion, the host portion, and the subnet mask. + - • Explain the characteristics and uses of the unicast, broadcast and multicast IPv4 addresses. • Explain public, private, and reserved IPv4 addresses. • Explain how subnetting segments a network to enable better	Multi-media Projector, Lecture notes, Tutorials, Pictorials	Convert binary numbers to decimal numbers Convert decimal numbers to Hexadecimal numbers	Demonstrate how to convert binary numbers to decimal numbers Demonstrate the conversion of decimal numbers to Hexadecimal numbers	Explain the structure of an IPv4 address including the network portion, the host portion, and the subnet mask.

	enable better communication. 5.7 Compute IPv4 subnets for a /24 prefix. 5.8 Describe the need for IPv6 addressing. 5.9 Describe how IPv6 addresses are represented. 5.10 Contrast types of IPv6 network addresses. 5.11 Describe how to configure static global unicast and link-local IPv6 network addresses. 5.12 Describe how to configure global unicast addresses dynamically.	communication. • Calculate IPv4 subnets for a /24 prefix. • Explain the need for IPv6 addressing. • Explain how IPv6 addresses are represented. • Compare types of IPv6 network addresses. • Explain how to configure static global unicast and link-local IPv6 network addresses. • Explain how to configure global unicast addresses dynamically.				
General Objective 6.0: Understand Basic router and switch configuration						
11-12	6.1 Explain how to access IOS or VRP device for configuration purposes.	• Explain how to access IOS or VRP device for configuration purposes.	Multimedia Projector, Routers and Switches, Packet Tracer or	• Configure a Switch and navigate through the IOS or VRP	Guide students to: • Navigate the IOS or VRP by Using Tera Term for Console	Explain how to navigate IOS or VRP to configure network devices.

	6.2 Explain how to navigate IOS or VRP to configure network devices. 6.3 Explain the command structure of IOS or VRP software. 6.4 Explain IOS or VRP device using CLI. 6.5 Explain IOS or VRP commands to save the running configuration. 6.6 Explain how devices communicate across network media. 6.7 Explain a host device with an IP address. 6.8 Explain connectivity between two end	• Explain how to navigate IOS or VRP to configure network devices. • Explain the command structure of IOS or VRP software. • Explain IOS or VRP device using CLI. • Explain IOS or VRP commands to save the running configuration. • Explain how devices communicate across network media. • Explain a host device with an IP address. • Explain connectivity between two end	eNSP or GNS3, Laptop Internet Books, Journals, Marker board, Marker, Textbooks, Tera Term, Putty	• Configure Initial • Switch Settings	Connectivity • Configure Initial • Switch Settings	
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	devices. 6.9 Explain initial settings on an IOS or VRP router. 6.10 Explain two active interfaces on IOS or VRP router. 6.11 Explain how devices use the default gateway.	devices. • Explain initial settings on an IOS or VRP router. • Explain two active interfaces on IOS or VRP router. • Explain how devices use the default gateway.				
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General Objective 7.0: Understand Network Security fundamentals

13-14	7.1 Explain why basic security measures are necessary on network devices. 7.2 List general security mitigation techniques. 7.3 Explain network devices with device hardening features to mitigate security threats.	• Explain why basic security measures are necessary on network devices. • • List general mitigation techniques. • Explain network devices with device hardening features to mitigate security threats.	Multimedia Projector, Routers and Switches, Packet Tracer or eNSP or GNS3, Laptop Internet Books, Journals, Marker board, Marker, Textbooks, TeraTerm, Putty	• Configure secure passwords and SSH on network devices.	Guide the students to configure secure passwords and SSH on network devices.	List general mitigation techniques
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General Objective 8.0: Know how to build small networks						
14-15	8.1 Explain the devices used in a small network. 8.2 Explain the protocols and applications used in a small network. 8.3 Explain how a small network serves as the basis of larger networks. 8.4 Explain common network troubleshooting methodologies. 8.5 List common connectivity problems associated with network devices	• Explain the devices used in a small network. • Explain the protocols and applications used in a small network. • Explain how a small network serves as the basis of larger networks. • Explain common network troubleshooting methodologies. • Explain common connectivity problems associated with network	Multimedia Projector, Routers and Switches, Packet Tracer or eNSP or GNS3, Laptop Internet Books, Journals, Marker board, Marker, Textbooks, Tera Term, Putty	• Design a small network, and use ping and traceroute to test for latency and connectivity. • Troubleshoot connectivity issues.	Guide the students to: • Design a small network, and use ping and traceroute to test for latency and connectivity. • Troubleshoot connectivity issues.	Explain the protocol and application used in a small network

VIRTUALIZATION AND CLOUD MIGRATION

PROGRAMME: HIGHER NATIONAL DIPLOMA IN NETWORKING AND CLOUD COMPUTING		
COURSE: VIRTUALIZATION AND CLOUD MIGRATION	CODE: NCC 312	CONTACT HOURS: 45 HOURS/WEEK
YEAR: 1 SEMESTER 1	Credit Units: 3	THEORETICAL: 1 HOURS/WEEK PRACTICAL: 2 HOURS/WEEK
GOAL: This course is designed to equip students with theoretical and practical skills required to understand virtualization technologies		
Course Specification: THEORETICAL CONTENT		PRACTICAL CONTENT
GENERAL OBJECTIVES: On completion of this course, the students should be able to: 1.0: Understand the concepts of virtualization. 2.0 Understand the local virtualization Technologies. 3.0 Know Cloud Service Providers. 4.0 Understand cloud computing and virtualization. 5.0 Understand virtualization management and security. 6.0 Understand virtualization trends and Emerging technologies		

PROGRAMME: HIGHER NATIONAL DIPLOMA IN NETWORKING AND CLOUD COMPUTING						
COURSE: VIRTUALIZATION AND CLOUD MIGRATION			CODE: NCC 312		CONTACT HOURS: 45 HOURS/WEEK	
YEAR: 1 SEMESTER 1			Credit Units: 3		THEORETICAL: 1 HOURS/WEEK PRACTICAL: 2 HOURS/WEEK	
GOAL: This course is design to equip students with theoretical and practical skills required to understand virtualization technologies						
Course Specification: THEORETICAL CONTENT				PRACTICAL CONTENT		
GENERAL OBJECTIVE 1.0: Understand the concepts of virtualization.						
Wee k	Specific Learning Outcome	Teachers’ Activities	Learning Resources	Specific Learning Outcome	Teachers’ Activities	Evaluation
GENERAL OBJECTIVE 2.0: Understand the Local Virtualization Technologies.						
2-3	2.1 Define hypervisor. 2.2 Identify their roles in local virtualization. 2.3 Describe the difference between type 1 and type 2 hypervisors. 2.4. Describe the features and capabilities of	• Explain hypervisors. • Explain the difference between type 1 and type 2 hypervisors. • Explain the features and capabilities of popular bare-metal hypervisors, using examples such Vmware vSphere/ESXi, Microsoft Hyper-V, etc	Internet, Local directory for copying the executable file, Markers, whiteboard, computer, Projector, Lecture notes,	Install any type 2 hypervisor, Create virtual machine instances, Guest OS installation and configuration, Setup a network and connect two or more OSs running on	Guide students to install any type 2 hypervisor, Guide students to create virtual machines, configure the VMs, install OSs, and manage the VMs and the OSs	Define hypervisor.

	popular local virtualization platforms.	Explain the features and capabilities of popular hosted hypervisors, using examples such as Oracle Virtual Box, Vmware Workstation, etc	Pictorials, PowerPoint slides, etc.	different virtual machines, Create snapshots, clones, and manage the VM.		
GENERAL OBJECTIVE 3.0: Know Cloud Service Providers.						
4-7	3.1 Explain major cloud service providers using virtualization 3.2 Describe features, pricing models, and use cases of different providers. 3.3 Explain how to use proxmox for virtualization	- Explain AWS virtual services features, pricing models and use cases - Explain Microsoft Azure virtual offerings features, pricing models and use cases. - Explain Google Cloud Virtual Solution features, pricing models and use cases - Explain how to use proxmox for virtualizations	Internet, Amazon EC2, Azure VM, or Google Compute Engine, Multi-media Projector, Lecture notes, Marker, Whiteboard, Computer, PowerPoint slides, etc.	- Setup 2 servers and install hypervisor type 1 - Prepare the servers by enabling virtualization in the BIOS set up - Install proxmox or any other Hypervisor Type 1 software - Set up networking On server 2 only: - Set up shared storage on server - Configure cluster - Create virtual machines	Guide students to: - Setup 2 servers and install hypervisor type 1 - Prepare the servers by enabling virtualization in the BIOS set up - Install proxmox or any other Hypervisor Type 1 software - Set up networking On server 2 only: - Set up shared storage on server	Describe features, pricing models, and use cases of different providers.

				• Manage virtual machines • Configure high availability and backup • Install OS on virtual machines • Select cloud service provider and setup a free account • Explore the providers features and pricing models	• Configure cluster • Create virtual machines • Manage virtual machines • Configure high availability and backup • Install OS on virtual machines Guide the students to select a cloud service provider like AWS, Azure, Google Cloud, etc, and setup an account Guide the students to explore the features and pricing model of the provider services.	
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GENERAL OBJECTIVE 4.0: Understand Cloud Computing and virtualization.						
8-9	4.1 Explain the relationship between cloud computing and virtualization. 4.2 Explain the cloud service models and their virtualization components. 4.3 Explain the benefits of virtualization in enabling cloud services.	• Explain cloud computing and how virtualization technologies is used in cloud computing. • Explain the roles of virtualization in cloud service models • Explain the impact of virtualization on cloud scalability.	Internet, Amazon EC2, Azure VM, or Google Compute Engine, Projector, Lecture notes, Marker, Whiteboard, Computer, PowerPoint slides, etc.	Launch a virtual server instance on the created account and configure instance details, Configure network and security group, Setup key pairs and access the server remotely, Assign Elastic IP and attach additional storage,	Guide students to launch a virtual server and configure details like OS, storage, etc, Guide students to configure network and security group in AWS VPC or Azure VN, Guide students to setup key pairs (SSH key pair) and access the server remotely, Guide students to assign Elastic IP and attach and manage additional storage like Amazon EBS or Azure Managed Disks	Explain the cloud service models and their virtualization components.

GENERAL OBJECTIVE 5.0: Understand Virtualization Management and Security.

10-11	5.1 Explain the tools for Managing and monitoring virtualized environments. Explain security considerations for virtual machines. 5.3 Explain the importance of patch management and resource optimization.	• Explain virtualization management tools • Explain security considerations for virtual machines such as guest OS hardening and network security • Explain optimization and patch management resource • Explain security best practices for virtualized environment	Virtualization management tools like vCenter, Hyper-V Manager, etc, Internet, Projector, Lecture notes, Marker, Whiteboard, Computer, PowerPoint slides, etc.	Explore different management tools for virtualized environment, Setup a virtualization management tools, Monitor virtualized resources, Configure security and hardening features, Explore automation and reporting features of the management tools	Guide students to explore and manage virtualization management tools Guide students to select the right tools for their created virtual environment, Guide students to explore the dashboard and features of the chosen tools to monitor resources like CPU, RAM, and storage usage, Guide students to investigate automation options such scripting task, automatic backups, etc.	Explain security best practices for virtualized environment
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GENERAL OBJECTIVE 6.0: Understand Virtualization Trends and Emerging Technologies.						
12-15	6.1 Explain how to identify the latest trends in virtualization and cloud technologies. 6.2 Explain how to identify the emerging virtualization cloud technologies. 6.3 Explain the role of artificial intelligence (AI) in optimizing virtualized environments. 6.4 Explain the importance of disaster recovery planning.	• Explain server less computing and Event-Driven Architectures • Explain Edge computing and its impact on virtualization • Explain AI-driven optimization in virtualized environments. • Explain common causes of IT disaster such hardware failure, natural disaster, etc • Explain the components of disaster recovery plan such as backup, replication, fail-over, etc • Explain the importance of Recovery Point Objective (RPO) and	Internet, Projector, Lecture notes, Marker, Whiteboard, Computer, PowerPoint slides, etc.	Configure regular data backups for virtualized resources, Test backup and restoration processes for virtual machines, Configure replication and fail-over to a secondary site or cloud region for disaster recovery, Test fail-over procedure, Test disaster recovery procedures and assess their effectiveness	Guide students on how to choose backup solutions offered by virtualization platform or third-party tools, Guide students to configure backup schedules and retention policies, test backup and restoration processes, Guide students to understand replication options within the virtualization platform (example Vmware SRM), Guide the students to test the configured fail-over procedure to ensure smooth	Explain AI-driven optimization in virtualized environments

		Recovery Time Objective (RTO)			transition to the secondary environment, Guide the students to simulate a disaster scenario in a controlled environment, Validate the recovery of virtual machines and applications	
ASSESSMENT: The continuous assessment; tests, quizzes will be awarded 40% of the total score. The end of the Semester Examination will make up for the remaining 60% of the total score.						

OPERATING SYSTEM FOR NETWORK ENVIRONMENT

PROGRAMME: HIGHER NATIONAL DIPLOMA (HND) NETWORKING AND CLOUD COMPUTING		
COURSE: OPERATING SYSTEM FOR NETWORK ENVIRONMENT	COURSE CODE: NCC 313	Contact Hours: 3 hours / week
	CREDIT UNIT: 3	Theoretical: 1 hour /week
Year: 1 Semester: 1	Pre-requisite:	Practical: 2 hours / week
GOAL: This course is designed to acquaint students with knowledge and skills in Operating System for Network Environment		
GENERAL OBJECTIVES: General Objectives: On completion of this course the student should be able to: 1.0 Understand Operating Systems. 2.0 Understand Linux Operating System 3.0 Know User and File Management 4.0 Understand Process Management 5.0 Comprehend Memory Management and Virtualization 6.0 Understand Network Addressing in Linux		

PROGRAMME: HIGHER NATIONAL DIPLOMA (HND) NETWORKING AND CLOUD COMPUTING						
COURSE: OPERATING SYSTEM FOR NETWORK ENVIRONMENT				COURSE CODE: NCC 313		Contact Hours: 3 hours / week
				CREDIT UNIT: 3		Theoretical: 1 hour / week
Year: I Semester: I		Pre-requisite:			Practical: 2 hours / week	
COURSE SPECIFICATION: THEORETICAL AND PRACTICAL						
GOAL: This course is designed to acquaint students with knowledge and skills in Operating System for Network Environment						
GENERAL OBJECTIVE 1.0: Understand Operating Systems						
THEORETICAL CONTENT				PRACTICAL CONTENT		
Week	Specific Learning Outcome	Teacher’s Activities	Resources	Specific Learning Outcome	Teacher’s Activities	Evaluation
1 – 2	1.1 Define Operating System(OS) 1.2 Explain the Evolution of OS 1.3 Describe the concept of OS in a Networking Environment 1.4 Describe mode of	- Explain operating System (OS) and its Evolution. - Describe the concept of OS in Networking Environment eg: Resource Management, Facilitate communication, Security and Authentication, File and Print Sharing, Network services, Resource Management, Remote access and control, Importance and features of OS, etc. - Describe mode of	Internet PC Loaded with virtualization software with different OS installed Presentation package Multimedia Projector Text Books Lecture Notes Marker Marker Board	- Create and manage user Account - Create, Delete, Copy and Moving Files and Directories using Windows OS - Use Task Manager / Activity Monitor to view and monitor system resources, running processes etc	Guide Students on how to: - Create, modify, and delete user accounts. Understand user privileges and experiment with different user roles. - Create, rename, move, and delete files and folders on your Windows operating system. Explore different file	List types of operating systems

	operation and the architecture of OS 1.5 Explain the types and classes of Operating System.	operation and the Architecture of OS ○ (Monolithic, Micro-kernel, ○ Layered Kernel) • Classify OS into batch, real time, timesharing and networking • Describe the types of OS such as: Microsoft Windows, Apple MacOS, Linux: Linux Distribution Red hat and Embedded Systems		• Perform basic Command-Line Interface (CLI) operations	attributes and permissions. • Use the built-in task manager (Windows) or Activity Monitor (macOS/Linux) to monitor system resources, view running processes, and terminate unresponsive applications. • Practice basic command-line operations. Navigate directories, list files, copy, move, and delete files using commands like ls, cd, cp, mv, and rm.	
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GENERAL OBJECTIVE: 2.0 Understand Linux Operating System

2 – 4	2.1 Define Linux Operating System 2.2 Give brief history and evolution of Linux 2.3 Describe the Open Source Philosophy and the GNU General Public License (GPL) 2.4 Describe Linux Distributions and Package Management System 2.5 Explain Installation Process (dual-boot or Virtual Machine set up) 2.6 Describe the basic Linux commands and command-line interface (CLI) 2.7 Describe File System hierarchy and Navigation 2.8 Explain the usage of “Getting Help”	• Explain Linux Operating System • Describe the Open Source Philosophy such as: Free Redistribution, Access to Source Code, Technology Neutrality, Transparency etc • Explain Linux Distribution and Package Management • Describe the Installation Process and the basic Linux Command-liner Interface • Show the basic Linux commands and command-line • Explain the File System hierarchy and Navigation,	Computer System Internet PC Loaded with virtualization software with different OS installed Presentation package Multimedia Projector Linux Distribution (Distro) eg Ubuntu, Fedora etc., Installation Media, Text books Marker/Marker Board	• Utilize a package manager to install and update software • Diagnose and address issues with package repositories and apply package management commands effectively • Set up a basic Linux networking environment • Experiment with help-related commands • Explore commands to search for commands based on keywords etc.	Guide Students on how to: • Utilize a package manager to install and update software • Apply package management commands effectively • Set up a basic Linux networking environment • Experiment with help-related commands like ‘info’, which provides comprehensive documentation for some commands and topics. • Use ‘apropos’ and ‘whatis’	Explain Installation Process (dual-boot or Virtual Machine set up)
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		Describe “Getting Help” eg: Introduction, Man Pages, Finding Commands and Documentation, Info Documentation etc			commands to search for commands based on keywords. Use the help’ option with various commands to access built-in command-line documentation	
GENERAL OBJECTIVE: 3.0 Know User and File Management						
5 - 6	3.1 Explain user accounts and groups 3.2 Explain user authentication, Password Policies, file permissions and ownership.	- Describe user account such as: Username, Password, Home Directory, Access Privileges, Personal Settings and Resource Usage. User Groups such as: Primary Group, Secondary Groups, Access Control etc. - Explain User authentication, Password Policies, File permissions and ownership - Describe File Manipulation and basic text editing such as: Copying and Moving	Computer Hardware, Linux Distribution (Distro) e.g. Ubuntu, Fedora etc., Installation Media, Marker Marker Board Text books	- Configure user authentication and permissions - Implement RAID configurations - Manage networked storage using Linux tools - Create, Delete, Copy and Move Files and Directories - Use Text Editors to Navigate, Move,	Guide Students on how to: - Configure user authentication and permissions - Implement RAID configurations - Manage networked storage using Linux tools - Create, rename, move, and delete files and	Explain user authentication, Password Policies, file permissions and ownership

	3.3 Explain the File Manipulation and basic text editing	Files, Changing Permissions, Formatting and Indentation, Inserting and Deleting Text, etc.		Search, and Replace • Create and Edit Scripts • Set up file sharing between computers	folders on your operating system. Explore different file attributes and permissions. • Use Text Editors to Navigate and Move, Search and Replace • Create and Edit Scripts • Set up file sharing between two computers using protocols like SMB (Server Message Block) or NFS (Network File System). Share a folder from one system and access it from another.	
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GENERAL OBJECTIVE: 4.0: Understand Process Management	
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7 – 9	4.1 Explain processes and threads 4.2 Explain Process states and management commands. 4.3 Explain Process Scheduling and Priorities, Process Creation and Termination, Concurrency and Parallelism and Inter-process communication (IPC) 4.4 Explain Deadlock Handling, Load Balancing and Resource Management	- Describe Processes and threads, - Process states and Management command, Process Scheduling considering factors like: process priority, communication delays, and fairness across networked machines. - Describe Concurrency, Parallelism, Process Creation and Termination, Inter Process Communication etc. - Explain the concept of Deadlock e.g. Resource Allocation Graph (RAG), Deadlock Detection, Timeouts and Aborts, Resource Pre-emption and Distributed Deadlock Detection.	Multimedia Projector Textbooks, Online Articles And Tutorials. Video Lectures And Courses. Computer System Marker Marker board	Monitor and control processes using Linux and Windows tools Demonstrate how OS schedules and switches between threads in a multithreaded applications Create, terminate, and monitor processes.	Guide Student how to: Demonstrate how OS schedules and switches between threads in a multithreaded applications Monitor and control processes using Linux/ Windows tools Create, terminate, and monitor processes;	Explain processes and threads
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	4.5 Explain Security and Isolation	Explain the shared responsibility model for cloud security				
General Objective 5.0: Comprehend Memory Management and Virtualization						
10 - 12	1.1 Explain memory Management 1.2 Explain Virtual memory and Paging 5.3 Explain Swap space Configuration 5.4 Describe Virtualization (Virtual Box, KVM)	- Describe Memory Management with respect to: Memory Hierarchy, Memory Allocation, Memory Protection, Virtual Memory, Swapping, Paging, Demand Paging, Page Replacement Algorithms and Fragmentation. - Describe the concept of virtualization	Open-Source Operating Systems Multimedia Projector Textbooks Online Articles And Tutorials. Video Lectures And Courses. Marker Marker Board •	- Configure memory settings and memory management strategies in a Linux and windows - Configure virtual memory settings, monitor paging behaviour, and optimize memory usage - Create virtual machines using virtualisation tools	Guide Students how to: - Configure memory settings and memory management strategies in a Linux and windows - Configure virtual memory settings, monitor paging behaviour, and optimize memory usage - Create virtual machines using virtualisation	Explain Virtual memory and Paging

					tools • Create and analyse Scenarios demonstrating deadlock and apply strategies to prevent and resolve deadlocks	
GENERAL OBJECTIVE: 6.0 Understand Network Addressing in Linux						
13 - 15	6.1 Explain Networking Protocols and Stacks 6.2 Explain IP Addressing and Subnetting 6.3 Explain Sockets, Socket APIs, Network Utilities and Commands 6.4 Describe Firewalls and Security2	• Describe Physical Layer, Data Link Layer, Network Layer, Transport Layer and Application Layer • Describe IP Addressing, IPv4 (Internet Protocol Version 4), IPv6 (Internet Protocol version 6), Subnetting etc. • Describe Network Utilities and Commands. • Explain the Key Functions of Firewalls, Types of Firewalls, Security Mechanisms,	Computer system, Internet connectivity LAN Multimedia Projector Textbooks, Online Articles Cheat Sheets and Quick References Practical Labs and Simulators.	Capture and analyze network traffic Configure a Linux computer to obtain IP details Configure IP addresses, subnetting, and routing Use network utilities and commands to diagnose, troubleshoot, and manage network connectivity issues	Guide Students on how to: Configure a Linux computer to obtain IP details Configure IP addresses, subnetting, and routing Use network utilities and commands to diagnose, troubleshoot, and manage network connectivity issues	Explain Network Monitoring and Analysis

		How Firewalls Work, Firewall Rules, Importance of Firewalls in Security				
	6.5 Explain Network Monitoring and Analysis	• Describe Network Monitoring and Analysis				

WIRELESS COMMUNICATION SYSTEMS

PROGRAMME: HIGHERE NATIONAL DIPLOMA(HND) NETWORKING AND CLOUD COMPUTING		
COURSE: WIRELESS COMMUNICATION SYSTEMS	COURSE CODE: NCC 314	Contact Hours: 3 Hours /week
	CREDIT UNIT: 3	Theoretical: 2 Hours /week
Year: 1 Semester: 1	Pre-requisite:	Practical: 1 Hour /week
GOAL: This course is designed to equip the students with theoretical and practical skills in wireless communication systems.		
GENERAL OBJECTIVES: On completion of the course, the student should be able to: 1.0 Understand wireless communication systems. 2.0 Understand mobile communication technologies. 3.0 Understand mobile Radio Propagation. 4.0 Understand the practical applications of mobile communications in various industries. 5.0 Know WLAN standards and the development of Wi-Fi technology 6.0 Understand WLAN technologies.		

PROGRAMME: HIGHER NATIONAL DIPLOMA (HND) NETWORKING AND CLOUD COMPUTING							
COURSE: WIRELESS COMMUNICATION SYSTEMS		COURSE CODE: NCC 314			Contact Hours: 3 Hours/Week		
		CREDIT UNIT: 3			Theoretical: 2 Hours/Week		
Year: 1 Semester: 1		Pre-requisite:			Practical: 1 Hour/Week		
GOAL This course is designed to equip the students with theoretical and practical skills in wireless communication systems.							
GENERAL OBJECTIVE 1.0: Know wireless communication systems.							
THEORETICAL CONTENT				PRACTICAL CONTENT			
Week	Specific Learning Outcome	Teacher’s Activities	Resources	Specific Learning Outcome	Teacher’s Activities	Evaluation	
1-3	1.1 Define wireless communication systems and their significance 1.2 Define components of wireless communication systems 1.3 Differentiate wireless communication systems with wired communication systems 1.4 Differentiate between analogue and digital communication systems 1.5 List types of wireless communication systems 1.6 Identify various applications of	• Explain what is wireless communication • Explain the components of wireless communication systems • Explain the difference between analogue and digital communication systems • Explain wireless communication systems • Explain wireless communication networks • Explain various	Textbooks, Multi-media Projector, Computer, Marker, Marker board, Lecture notes, Animation Videos/illustrations , walkie-talkie, Oscilloscope.	• Demonstrate wireless communication is achieved using walkie-talkies • Identify the various components that make up the systems • Use a simple oscilloscope or waveform visualization software to demonstrate the difference between analogue and digital signals.	• Guide students to set up a simple wireless communication system using two devices like walkie-talkies or wireless modules. • Show the components, such as transmitters, receivers, antennas, and how they interact. • Guide students to use a simple oscilloscope or waveform visualization software to show them the difference between analogue and digital signals. • Demonstrate how noise affects analogue signals	Define wireless communication systems and their significance	

	wireless communication systems	applications of wireless communication systems		Demonstrate how noise affects analogue signals compared to digital ones	compared to digital ones.	
GENERAL OBJECTIVE: 2.0 Understand Mobile Communication Technologies.						
4-5	2.1 Define mobile communication 2.2 2.3 List the different generations of mobile communication 2.4 List the applications of mobile communication 2.5 Describe cellular network architecture and its components 2.6 Outline the evolution of mobile generations (1G to 5G) 2.7 2.8 List key features and benefits of each mobile generation.	• Explain what is mobile communication • Explain the different generation of mobile communication • Explain the applications of mobile communication • Explain cellular network architecture and its constituent elements. • Demonstrate how a cellular telephone call is made. • Explain the unique characteristics and improvements introduced in each mobile generation	Textbooks, Multi-media Projector, Computer, Marker, Marker board, Lecture notes, Animation Videos, image illustrations, mobile phone, walkie-talkie.	• Implement a simple mobile communication setup using mobile phone to show how information is transmitted and received • Show how cellular network using a mobile phone	• Guide the students on how to setup a simple mobile communication using devices like walkie-talkies or two-way radios. Explain how information is transmitted and received. • Set up a mock cellular network using simple equipment or visual aids. • Show the components such as base stations, cell towers, mobile devices, and the communication flow between them	List the different generations of mobile communication

GENERAL OBJECTIVE 3.0 Understand mobile Radio Propagation						
6 – 7	3.1 Define the concept of Radio wave propagation 3.2 Define free space propagation model 3.3 Outline the three basic Propagation mechanism 3.4 Explain ground reflection (Two-Ray) Model 3.5 Explain indoor propagation models 3.6 Explain Outdoor propagation models 3.7 Describe Practical Link Budget design using path loss models 3.8 Define Multiple access technique for wireless communications	• Explain the concept of Radio wave propagation • Describe free space propagation model • Explain the three basic Propagation mechanism • Describe ground reflection (Two-Ray) Model • Explain Indoor propagation models • Explain Outdoor propagation models • Explain practical Link Budget design using path loss models • Explain Multiple access technique for wireless communications	Textbooks, Multi-media Projector, Computer, Marker, Marker board, Lecture notes, Animation Videos/illustrations.			Describe free space propagation model
GENERAL OBJECTIVE: 4.0: Comprehend the practical applications of mobile communications in various industries						
7-8	4.1 Define mobile communication applications in transportation and logistics.	• Discuss case studies and examples Highlight the use of mobile communications in different sectors.	Textbooks, Multi-media Projector, Lecture notes, Animation Videos/illustrations			Explain the use of mobile communications in agriculture and rural development.

	4.2 Explain the use of mobile communications in agriculture and rural development. 4.3 Explain the impact of mobile communications on banking and finance.	Facilitate a discussion on the benefits and challenges of applying mobile communications in diverse industries.				
GENERAL OBJECTIVE: 5.0 Understand WLAN standards and the development of Wi-Fi technology.						
9 - 11	5.1 Define the concept of WLAN (Wireless Local Area Network) and its importance 5.2 Describe the evolution of Wi-Fi standards (802.11a/b/g/n/ac/ax) 5.3 Differentiate between various Wi-Fi frequencies and their applications.	- Introduce the concept and significance of WLANs. - Explain timeline of Wi-Fi standards and their respective enhancements. - Discuss the utilization and advantages of different Wi-Fi frequencies.	Textbooks, Multi-media Projector, Lecture notes, Animation Videos/illustration	Demonstrate different Wi-Fi technologies using mobile hotspot	Guide students to demonstrate different Wi-Fi technologies using mobile hotspot	Differentiate between various Wi-Fi frequencies and their applications
GENERAL OBJECTIVE: 6.0 Understand WLAN technologies.						
12- 15	6.1 Explain the importance of standardization in WLAN technology. 6.2 List common WLAN technical standards and regulations.	Explain the components and functioning of	Textbooks, Multi-media Projector, Lecture notes, Animation Videos/illustration s, marker,			Describe the basic components of a WLAN

	6.3 Describe the basic components of a WLAN 6.4 Explain the principles of WLAN communication 6.5 Outline the architecture of IEEE 802.11 protocol 6.6 Differentiate the modes and frame types in IEEE 802.11 protocol	• Describe the basic components of a WLAN Discussions on the principles of wireless communication within WLANs. • Describe the architecture and workings of the IEEE 802.11 protocol. • Explain the various modes and types of frames used in IEEE 802.11 protocol.	Markerboard,				
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COMPUTER ARCHITECTURE

PROGRAMME: HIGHER NATIONAL DIPLOMA (HND) NETWORKING AND CLOUD COMPUTING		
COURSE: COMPUTER ARCHITECTURE	COURSE CODE: NCC 315	Contact Hours: 3 hours /week
	CREDIT UNIT: 3	Theoretical: 2 hours /week
Year: 1 Semester: 1	Pre-requisite:	Practical: 1 hour /week
GOAL: This course is designed to provide students with knowledge of Internal Structure and Functions of the Computer System		
GENERAL OBJECTIVES: On completion of the course, the student should be able to: 1.0 Understand the Concept of Computer Organisation and Architecture 2.0 Understand Analysis of Different Computer System Architectures. 3.0 Know Functional Units and Operations 4.0 Understand Memory/Storage Organisation and Management 5.0 Comprehend Instruction Level Parallelism and Parallel Computers 6.0 Understand Multiprocessors and Thread-Level Parallelism		

PROGRAMME: HIGHER NATIONAL DIPLOMA NETWORKING AND CLOUD COMPUTING						
COURSE: COMPUTER ARCHITECTURE		COURSE CODE: NCC 315			Contact Hours: 3 hours /week	
		CREDIT UNIT: 3			Theoretical: 2 hours /week	
Year: I Semester: I		Pre-requisite:			Practical: 1 hours /week	
GOAL: This course is designed to provide the student with a thorough understanding of the key concepts and principles of computer architecture						
GENERAL OBJECTIVE 1.0: Understand the Concept of Computer Organisation and Architecture						
THEORETICAL CONTENT				PRACTICAL CONTENT		
Week	Specific Learning Outcome	Teacher's Activities	Resources	Specific Learning Outcome	Teacher's Activities	Evaluation
1-3	1.1 Explain the concept of computer organisation and architecture 1.2 Explain the three major components of a computer and its functions 1.3 Describe the internal structures of the CPU and its functions	- Explain the theoretical concept of computer organisation and architecture, show the difference between the two with examples - Explain the three major components of a computer system with the functions, including the CPU, memory, and I/O - Describe the structure and	Marker board, Markers, Documentary films, Multimedia Projector and Screen, Case studies, Power point slides, Internet, Textbooks, Computer, Flip charts etc.	Identify computer organisation and architecture Identify each major component on the motherboard, including microprocessor, BIOS and CMOS IC, cache, I/O ports and terminals	Use different component structural relationship and functional behaviour to illustrate the concepts of computer organisation and architecture Use a typical closed or complete motherboard to show the microprocessor, BIOS and CMOS IC, cache, I/O ports and other terminal and	Explain the concept of computer organisation, computer architecture and show the difference

	1.4 Describe systems evolution 1.5 Explain processor performance measurement metrics 1.6 Explain the factors affecting processor performance	Functioning of the control unit, execution unit like arithmetic logic unit, control unit and registers within the CPU. - Describe Intel and Advance RISC Machine (ARM) processors evolution - Explain the metrics used in measuring processor performance such as speed, power, and Throughput. - Explain the factors affecting processor		Draw the detailed diagram of a typical (Harvard or Von Neuman) architecture to show the internal layout of the CPU Identify different processors and their evolution Check the speed, power, and throughput of a processor Identify how to check clock speed, smart cache, turbo boost, hyper threading, UHD graphics features on a typical system	explain the function of each Use either Harvard or Von Neuman architecture diagram to show the CPU structure Use different processors to describe the evolution of the systems Guide students to check the speed, power, and throughput of a processor. Guide students to identify how to check clock speed,	Explain the metrics used in measuring processor performance such as speed, power, and throughput. Explain the factors affecting processor performance such as architecture design, clock speed,
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	performance such as architecture design, clock speed, instruction set complexity, and parallel processing capabilities • Discuss the concept of processor benchmarking and the use of Benchmarking 1.7 Explain the concept of processor benchmarking and the use of benchmarking software. 1.8 Explain Microprocessor - I/O Interfacing 1.9 Explain I/O interface (interrupt and DMA mode) serial and parallel communications.	Identify how microprocessor communicate with outside world Use method of interrupt and DMA for interfacing Use of different ports, terminal and connectors Use spooling in sending data from the system to a printer	smart cache, turbo boost, hyper threading, UHD graphics features on a typical system	instruction set complexity, and parallel processing Capabilities.
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GENERAL OBJECTIVE: 2.0 Understand Analysis of Different Computer System Architectures						
4-6	2.1 Explain types of computer architecture	Explain types of Computer architecture including Harvard and Von Neumann architecture	Marker board, Markers, Documentary films, Multimedia Projector and Screen, Case studies, Power point slides, Internet, Textbooks, Computer, Flip charts etc.	Disassemble and assemble system unit	Guide students to disassemble and assemble system unit	Explain Instruction Cycle
	2.2 Explain Instruction Cycle	Explain instruction cycle, including Fetch Decode and execute				
	2.3 Describe Registers Set	Describe different CPU registers, including general and specific purpose				
	2.4 Explain Instruction Set	Explain the use of various set of instruction in program development				
	2.5 Explain the instruction set architecture	Explain different instruction set architecture design, including				

		Reduced Instruction Set Computers, Complex Instruction set computers • Explain different addressing modes, including Immediate, direct, indirect and indexing • Explain different operations, including arithmetic, logic, data movement and control operations				
	2.6 Explain different addressing modes 2.7 Explain different types of operations					

● GENERAL OBJECTIVE 3.0: Know Functional Units and Operations						
7-8	3.1 Explain Arithmetic and logic unit	Explain basic arithmetic operations, including adder, subtractor, multiplier, two's complement addition	Marker board, Markers, Documentary films, Multimedia Projector and Screen, Case studies, Power point slides, Internet, Textbooks, Computer, Flip charts etc.	Identify system bus diagram of a motherboard	Guide student to identify system bus diagram of a motherboard	
	3.2 Explain Floating points units	Explain floating point arithmetic operations				
	3.3 Explain Graphical Processing Unit (GPU)	- Explain GPU				
	3.4 Explain control unit design	Explain hardwired and Micro programmed control unit, control unit design				
	3.5 Explain system bus	Explain control bus, address bus and data bus				
● GENERAL OBJECTIVE 4.0: Understand Memory/Storage Organisation and Management						
9-10	4.1 Explain the hierarchy of different types of memory and storage devices	Explain the hierarchy	Marker board, Markers, Documentary films,	Use simulation software to visualize how data is stored and	Guide students to use simulation software to	Explain the hierarchy of different types

		• and characteristics of • different types of • memory and storage • devices (Static RAM, Dynamic RAM, ROM, • Hard disk, SSD, Cache • etc.). levels of cache, cache replacement algorithm, write policy	Multimedia Projector and Screen, Case studies, Power point slides, Internet, Textbooks, Computer, Flip charts Simulink Stateflow etc.	retrieved in different memory types. Format HDD drive, both logically and physically, create partitions, create file systems NTFS, Linux file system(ext3) Demonstrate different speed of data or file transfer using different storage and memories Create Virtual memory	visualize how data is stored and retrieved in different memory types. Guide students on how to format HDD drive, both logically and physically, create partitions, create file systems Use USB flash, external HDD, SSD to copy file to demonstrate different speed of data or file transfer using different medium Guide students to create Virtual memory	of memory and storage devices
	4.2 Explain the characteristics of different types of memory and storage devices					
	4.3 Describe how data is stored and retrieved in different memory types	• Explain different memory characteristic, including location, capacity, access method, physical type, speed				
	4.4 Describe the performance differences among various	• Explain how data is stored and retrieved in different memory types.				

	types of memory and storage devices. 4.5 Describe the concept and the use of virtual memory in computer systems.	- Discuss performance differences among various types of memory and storage devices. - Discuss the use of virtual memory in computer systems. - Explain the concept of memory management in operating systems, including line replacement strategies				
• GENERAL OBJECTIVE: 5.0 Comprehend Instruction Level Parallelism and Parallel Computers						
11-12	5.1 Explain the concept of Instruction Level Parallelism (ILP) and its importance in increasing computing performance.	Explain the concept of Instruction Level Parallelism (ILP) and its importance in increasing computing performance.	Marker board, Markers, Documentary films, Multimedia Projector and Screen, Case studies, Power point slides, Internet, Textbooks, Computer, Flip charts Simulink, etc.	Use different techniques to increase ILP using simulation software. Use simulation software to explore the concepts of out-of-order execution and speculative execution	Guide students to use different techniques to increase ILP using simulation software. Assist students to use simulation software to explore the concepts of out-of-order execution	Explain the impact of ILP on pipeline performance. Describe the concepts of out-of-order execution and speculative

	5.2 Explain hardware and software techniques used to increase ILP.	• Discuss the hardware and software techniques used to increase ILP.			and speculative execution.	execution in the context of ILP.
	5.3 Explain the impact of ILP on pipeline performance.	• Explain the impact of ILP on pipeline performance.				
	5.4 Describe the concepts of out-of-order execution and speculative execution in the context of ILP.	• Describe the concepts of out-of-order execution and speculative execution in the context of ILP.				
	5.5 Describe the implications of dependencies between instructions in the context of ILP.	• Describe the implications of dependencies between instructions in the context of ILP. • Explain Single Instruction, single				

	5.6 Explain Flynn's classification parallel computers	Data(SISD), Single Instruction Multiple Data(SIMD), Multiple Instruction Single Data(MISD), Multiple Instruction Multiple Data(MIMD)				
GENERAL OBJECTIVE: 6.0 Understand Multiprocessors and Thread-Level Parallelism						
13-15	6.1 Explain the basic concepts of multiprocessors and thread-level parallelism. 6.2 Explain the different architectures of multiprocessor systems 6.3 Explain the importance of cache coherence and memory	- Explain the basic concepts of multiprocessors and thread-level parallelism. - Discuss the different architectures of multiprocessor systems, such as symmetric multiprocessing (SMP) and massively parallel processing (MPP). - Explain the importance of cache coherence	Marker board, Markers, Documentary films, Multimedia Projector and Screen, Case studies, Power point slides, Internet, Textbooks, Computer, Flip charts Simulink Stateflow etc.	Use different architectures of multiprocessor systems, such as symmetric multiprocessing (SMP) and massively parallel processing (MPP). Carry out parallel programming, and exploit thread-level parallelism.	Assist students to use different architectures of multiprocessor systems, such as symmetric multiprocessing (SMP) and massively parallel processing (MPP). Guide students to carry out parallel programming, and exploit thread-level parallelism.	Explain the basic concepts of multiprocessors and thread-level parallelism. Discuss the different architectures of multiprocessor systems, such as symmetric multiprocessing (SMP) and massively parallel processing (MPP).

	consistency in multiprocessor systems. 6.4 Explain the role of parallel programming, and how to exploit thread-level parallelism. 6.5 Explain the performance implications and scalability of multiprocessor systems.	and memory consistency in multiprocessor systems. • Explain the role of parallel programming, and how to exploit thread-level parallelism. • Explain the performance implications and scalability of multiprocessor systems.				Explain the performance implications and scalability of multiprocessor systems.
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PYTHON PROGRAMMING BASICS

PROGRAMME: HIGHER NATIONAL DIPLOMA (HND) NETWORKING AND CLOUD COMPUTING		
COURSE: PYTHON PROGRAMMING BASICS	COURSE CODE: NCC 316	Contact Hours: 3 hours /week
	CREDIT UNIT: 3	Theoretical: 1 hour /week
Year: 1 Semester: 1	Pre-requisite:	Practical: 2 hours /week
GOAL: To enrich students with general understanding of Python to be able to create basic implementable projects		
GENERAL OBJECTIVES: On completion of this course, the student should be able to: 1.0 Understand Python Development Environment 3.0 Understand Datatypes and Arrays 4.0 Understand Control Structures 5.0 Understand Functions, Libraries and Modules 6.0 Understand String Operations 7.0 Understand List, Tuple, Dictionary and Set 8.0 Know The Concept Of Object Oriented Programme 9.0 Know how to work with Databases 10.0 Understand Basics of GUI Programming		

PROGRAMME: HIGHER NATIONAL DIPLOMA NETWORKING AND CLOUD COMPUTING							
COURSE: Python Programming Basics			COURSE CODE: NCC 316			Contact Hours: 2 Hours/week	
			CREDIT UNIT: 2			Theoretical: 1 Hour/week	
Year: I Semester: I			Pre-requisite:			Practical: 1 Hours/week	
GOAL: To enrich students with general understanding of Python to be able to create basic implementable projects							
GENERAL OBJECTIVE 1.0: Understand Python Development Environment							
THEORETICAL CONTENT				PRACTICAL CONTENT			
Week	Specific Learning Outcome	Teacher’s Activities	Resources	Specific Learning Outcome	Teacher’s Activities	Evaluation	
1	1.1 Explain the Installation of Python Interpreter and Running Simple Programs 1.2 Explain the main features of Python. 1.3 Explain the Application of Python 1.4 Explain the functions of the Python Interpreter	- Describe the features of Python; It is interpreted, Object Oriented, Interactive, and a Scripting language. - Explain the various application areas of Python - Explain the difference between an +interpreted language and a compiled language.	PC/Laptop, Multimedia projector, Marker, Markerboard, Python Interpreter, IDE - PyCharm. - IDLE - Sublime - Vim. - Atom. - Jupyter Notebook. - Eclipse - PyDev - GNU Emacs. Practical manual, Books	- Download and Install Python Interpreter and configure the running environment - Install an IDE and Run simple Python program - Familiarize with Python Program Development Environment. - Create python variables, and operators	Guide students to download and Install Python Interpreter and configure the running environment - Guide students to download Python Interpreter from www.python.org and install. - Guide students to install appropriate IDE such as IDLE, Pycham, Jupyter, Spyder, Sublime, PyDev.	What are the main features of Python programming language? What are the application areas of Python What is the function of Python IDE? What are the rules guiding Python	

	1.5 Explain Python development environment. 1.6 Explain how to download and install Python Interpreter and IDE 1.7 Explain Python variables and Operators	• Explain the functions and different kinds of Python development environment. • Discuss how to download and install Python Interpreter and IDE • Explain how to use Python variables and operators in python codes			Demonstrate to the students how to write python programs with variables and operators	Variable naming? Write a command line program to display “Hello Python”
GENERAL OBJECTIVE: 3.0 Understand Datatypes and Arrays						
3	3.1 Define Data types 3.2 Explain the various data categorizations - Data types in Python - Numeric - Dictionary - Boolean - Set - Sequence type - Etc.	• Explain the various Python data categorization • Explain Python array and how to	PC/Laptop, Multimedia projector, Marker, Markerboard, Python Interpreter, IDE • PyCharm. • IDLE • Sublime • Vim. • Atom. • Jupyter	• Use Python data types • Create and Manipulate arrays in Python • Create and manipulate List, Tuple and	• Guide the students to use Python data types. • Guide the students to create manipulate Arrays in Python • Guide the students to	What are the various Python data types? Mention four types of Arrays in Python Write Python codes to create store and

	3.3 Explain why and how Arrays are used in Python 3.4 Explain how to Create and manipulate Python Arrays	Use Arrays in Python • Discuss how to Create and manipulate Python Arrays	Notebook. • Eclipse • PyDev • GNU Emacs. Practical manual, Books.	Dictionary in Python	create and manipulate List, Tuple and Dictionary in Python	manipulate the content of an Array
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GENERAL OBJECTIVE 4.0: Understand Control Structures

4-5	4.1 Explain conditional statements in Python 4.2 Explain Loops in Python; For Loops, While Loops Explain the Range Function	• Describe the use of conditional blocks such as IF...ELIF and ELSE • Explain how FOR and WHILE loop constructs work. • Describe the Range Function	PC/Laptop, Multimedia projector, Marker, Markerboard, Python Interpreter, IDE • PyCharm. • IDLE • Sublime • Vim. • Atom. • Jupyter Notebook. • Eclipse • PyDev • GNU Emacs. Practical manual, Books	• Write and execute Python codes using IF statement, ELIF construct, and using IF...ELIF...ELSE construct. • Write programs that use the WHILE loop, WHILE with BREAK, WHILE with CONTINUE • Write programs that use the FOR loop, FOR with	• Demonstrate the implementation of Conditional Statements. • Guide students on using IF, ELIF, ELSE statements. Demonstrate the implementation of FOR and WHILE • Loops. Guide students on using WHILE, BREAK, and CONTINUE Commands.	Write a program that prints your name 100 times. Write a program to accept values, compare and print in specific order Use a for loop to print a triangle (of any character). Allow the user to specify how high the triangle should be.
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				BREAK		
GENERAL OBJECTIVE 5.0: Understand Functions, Libraries and Modules						
6-7	5.1 Explain Functions and differentiate between Library functions and User- defined functions. 5.2 List the rules for creating functions. 5.3 Explain the need for Recursive functions.-*-++ 5.4 Explain module	• Explain Functions and function parameters. • Outline the rules for creating functions • Explain how Recursive functions work • Explain modules • Explain how recursive functions work	PC/Laptop, Multimedia projector, Marker, Markerboard, Python Interpreter, IDE • PyCharm. • IDLE • Sublime • Vim. • Atom. • Jupyter Notebook. • Eclipse • PyDev • GNU Emacs. Practical manual, Books	• Write programs that calls built-in functions from the Python Library • Write a program that is made of at least two modules.	• Guide students to: • Use the Def command to define a function and set its parameters. Call the function. • Use the Return statement to exit a function • Create a recursive function	Explain Functions and function parameters. Write programs that calls built-in functions from the Python Library Write a program that is made of at least two modules

GENERAL OBJECTIVE: 6.0 Understand Strings operations						
8	6.1 Define strings 6.2 Explain string data types and string Methods 6.3 Explain how to use String Operators to Manipulates Strings 6.4 Identify and implement String Build-in Methods	• Explain String data type in Python • List the String Methods and Implementation • Identify and implement String Build-in Methods • I9+3 • 01n Methods	PC/Laptop, Multimedia projector, Marker, Markerboard, Python Interpreter, IDE • PyCharm. • IDLE • Sublime • Vim. • Atom. • Jupyter Notebook. • Eclipse • PyDev • GNU Emacs. Practical manual, Books	• Write codes to Create and Manipulate Strings • Use Operators to manipulate strings • Use build-in methods	Guide the students to: • Guide students to use Operators to manipulate strings • Guide students to use build-in methods	List the String Operators Write Python Code to Manipulate String with Build-in Methods
GENERAL OBJECTIVE 7.0: Understand List, Tuple, Dictionary and Set						
9-10	7.1 Describe List, Tuple, Dictionary and Set 7.2 Explain the Basic Operations of List, Tuple, Dictionary and Set - Indexing - Slicing	• Explain List, Tuple, Dictionary and Set • Explain basic List Operations	PC/Laptop, Multimedia projector, Marker, Markerboard, Python Interpreter, IDE • PyCharm. • IDLE	• Manipulate List, Tuple, Dictionary and Set in Python • Demonstrate Indexing, Slicing, and Matrixes in List and Tuple	Illustrate how to: • Create and Manipulate List • Create and Manipulate Tuple • Create and	What are the similarities and differences between List and Tuple? What are the similarities and

	- Matrixes 7.3 Explain the Built-in Functions and Methods	Explain Built-in List Functions and Methods	- Sublime - Vim. - Atom. - Jupyter Notebook. - Eclipse - PyDev - GNU Emacs. Practical manual, Books.		Manipulate Dictionary Create and Manipulate Set	differences between a dictionary and a set? State the properties of a dictionary
General Objective 8.0: Know the concepts of Object Oriented Programming						
11-12	8.1 Explain Object Oriented programming 8.2 Define Class and Object 8.3 Define Methods 8.4 Define Parent and Child Classes	- Explain Object Oriented concepts: Abstraction, Polymorphism, Inheritance, and Encapsulation - Explain class and object - Explain Methods and how they relate to Objects in a Class. - Explain Parent Class and Child Class	PC/Laptop, Multimedia projector, Marker, Markerboard, Python Interpreter, IDE - PyCharm. - IDLE - Sublime - Vim. - Atom. - Jupyter Notebook. - Eclipse - PyDev - GNU Emacs. Practical manual, Books	- Create a class named MyClass, with a property named x - Create an object named p1, and print the value of x - Create a class named Person, use the - init() function to assign values for name and age - Insert a function that prints a greeting, and execute it on the	- Demonstrate the concept of OOP in Python - Demonstrate how to create object and class	Define OOP Write a python code to create and object and print the value

				p1 object • Create a Parent Class named Person, with firstname and lastname properties, and a printname method • Create a class named Student, which will inherit the properties and methods from the Person class		
General Objective 9.0: Know how to work with Databases						
13-14	9.1 List the different databases that Python API supports - 9.2 Explain MySQL database commands.	• Explain the need for database • List and explain the different Databases that Python API supports. • Explain database operations and the syntaxes and functions of the following database command: Create Database, Create	PC/Laptop, Multimedia projector, Marker, Marker board, Python Interpreter, IDE • PyCharm. • IDLE • Sublime • Vim. • Atom. • Jupyter Notebook. • Eclipse • PyDev	• Create a database named "mydatabase" using the Create Database command. • Create a table named "customers" with two fields "Name" and "Address" in "mydatabase"	Guide students to: • Create a database named "mydatabase" using the Create Database command. Create a table named "customers" with two fields "Name" and	Use the appropriate SQL statement to create a Database with at least two tables and write statements to manage the content of the tables

		Table, Insert, Select, Where, Order by, Delete, Drop Table, Update, Join	• GNU Emacs. Practical manual, Books	using the Create Table command. • Insert a record in the "customers" table using the Insert Into command Query the "customers" using the Select..From..Where statement Delete a record from "customers" using the Delete...From...Where statement Delete "customers" from "mydatabase" using the Drop Table statement	"Address" in "mydatabase" using the Create Table command. • Insert a record in the "customers" table using the Insert Into command • Query the "customers" using the Select..From..Where statement • Delete a record from "customers" using the Delete...From...Where statement • Delete "customers" from "mydatabase" using the Drop	
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					Table statement	
General Objective 10: Understand uses of Graphic User Interface (GUI) Programming						
15	10.1 Explain GUI 10.2 Explain how to implement GUI Elements -Labels -Entry box -Grids -Buttons -Canvass 10.3 Explain the concept of Global variables 10.3 Explain callbacks and lambda	• Describe GUI imports modules • Explain how to create labels, text boxes, buttons, grids, canvass, and window colors	PC/Laptop, Multimedia projector, Marker, Markerboard, Python Interpreter, IDE • PyCharm. • IDLE • Sublime • Vim. • Atom. • Jupyter Notebook. • Eclipse • PyDev • GNU Emacs. Practical manual, Books	• Import GUI module • Create Labels • Creates a simple entry box using Entry () method and place it on the screen. • Use the get(), delete() and insert () methods to get input from the box, erase the content of the box and to insert text into the entry box	• Demonstrate how to import the modules for GUI. • Guide the students to create Labels, text boxes, buttons, grids for accepting user inputs and manipulate contents	Explain how to create labels, text boxes, buttons, grids, canvass, and window colors

YEAR ONE SEMESTER TWO COURSES

ROUTING AND SWITCHING 1

PROGRAMME: HIGHER NATIONAL DIPLOMA NETWORKING AND CLOUD COMPUTING		
COURSE: Routing and Switching 11	COURSE CODE: NCC 321	Contact Hours: 4 hours /week
	CREDIT UNIT: 4	Theoretical: 1 hour /week
Year: 1 Semester: 11	Pre-requisite: NCC 311	Practical: 3 hours /week
GOAL: This Course is designed to equip students with the essentials of Routing and Switching Technologies.		
GENERAL OBJECTIVES: On completion of the course, the student should be able to: 1.0 Understand Basic Device Configuration 2.0 Understand Virtual Local Area Networks (VLANs) 3.0 Understand Inter-VLAN Routing 4.0 Know Spanning Tree Protocol Concepts 5.0 Know link Aggregation Technology 6.0 Understand Dynamic Host Configuration Protocol for IPv4/ IPv6 7.0 Understand Stateless Address Auto Configuration (SLAAC)		

PROGRAMME: HIGHER NATIONAL DIPLOMA NETWORKING AND CLOUD COMPUTING						
COURSE: Routing and Switching1			COURSE CODE: NCC 321		Contact Hours: 4 hours /week	
			CREDIT UNIT: 4		Theoretical: 1 hours /week	
Year: I Semester: 11		Pre-requisite: NCC 311			Practical: 3 hours /week	
GOAL: This Course is designed to equip the students with the essentials of Switching and Routing Technologies.						
GENERAL OBJECTIVE 1.0: Understand Basic Device Configuration						
THEORETICAL CONTENT				PRACTICAL CONTENT		
Week	Specific Learning Outcome	Teacher’s Activities	Resources	Specific Learning Outcome	Teacher’s Activities	Evaluation
1-2	1.1 Define switch 1.2 Explain Frame Forwarding 1.3 Define Switching Domains 1.4 Explain how to setup a Switch with Initial Settings. 1.5 Define Switch Ports. 1.6 Explain Remote Access. 1.7 Explain Router Configuration.	- Explain Frame Forwarding. - Explain Switching Domains. - Configure a Switch with Initial Settings. - Explain Switch Ports. - Explain Secure Remote Access. - Explain Basic	Multimedia Projector, Routers and Switches, Packet Tracer or eNSP or GNS3, Laptop Internet Books, Journals, Marker board, Marker, Textbooks, TeraTerm, Putty	- Configure Switches, Cable the Network and Verify the Default Switch Configuration - Configure Basic Network Device Settings - Verify and Test Network Connectivity	Guide students to: - Configure Switches, Cable the Network and Verify the Default Switch Configuration - Configure Basic Network Device Settings - Verify and Test Network Connectivity	Define switch. Define frame forwarding. Define switching domains

	1.8 Explain Directly Connected Networks.	Router Configuration • Explain directly Connected Networks.				
GENERAL OBJECTIVE: 2.0 Understand Virtual Local Area Networks (VLANs)						
4 - 5	3.1 Explain the purpose of VLANs in a switched Network. 3.2 Explain frame forwarding within a Multi-Switched Environment using VLAN 3.3 Explain how to Setup a switch Port to be assigned to a VLAN. 3.4 Explain how to Configure a switch port for VLAN	• Explain the purpose of VLANs in a switched Network. • Explain frame forwarding within a Multi-Switched Environment using VLAN • Explain how to configure a switch Port to be assigned to a VLAN. • Explain how to Configure a switch port for VLAN Trunks • Explain how	Multimedia Projector, Routers and Switches, Packet Tracer or eNSP or GNS3, Laptop Internet Books, Journals, Marker board, Marker, Textbooks, TeraTerm, Putty	• Build a Network; • Create VLAN • Configure VLANs and Trunking • Configure Basic Device Settings • Assign Switch Ports • Maintain VLAN Port Assignments from the VLAN Database • Configure an 802.1Q Trunk Between the Switches	Guide students to: • Build a Network; • Configure VLANs and Trunking • Configure Basic Device Settings • Create VLANs; Assign Switch Ports • Maintain VLAN Port Assignments and the VLAN Database • Configure an 802.1Q Trunk Between the Switches	Explain the purpose of VLANs in a switched Network.

	Trunks 3.5 Explain how to Configure Dynamic Trunking Protocol	to Configure Dynamic Trunking Protocol				
GENERAL OBJECTIVE: 3.0 Understand Inter-VLAN Routing						
6 - 7	3.1 Explain options for configuring inter-VLAN routing. 3.2 Explain Router-on-a-stick inter-VLAN routing. 3.3 Explain inter-VLAN routing using Layer 3 switching. 3.4 3.5 Explain troubleshooting in inter-VLAN.	- Explain options for configuring inter-VLAN routing. - Explain Router-on-a-stick inter-VLAN routing. - Explain inter-VLAN routing using Layer 3 switching. - Explain troubleshooting in inter-VLAN.	Multimedia Projector, Routers and Switches, Packet Tracer or eNSP or GNS3, Laptop Internet Books, Journals, Marker board, Marker, Textbooks, TeraTerm, Putty	Configure Router-on-a-Stick Inter-VLAN Routing. Troubleshoot Inter-VLAN Routing. Configure Inter-VLAN Routing.	Demonstrate how to: Configure Router-on-a-Stick Inter-VLAN Routing. Troubleshoot Inter-VLAN Routing. Configure Inter-VLAN Routing	Explain options for configuring inter-VLAN routing

GENERAL OBJECTIVE: 4.0 Know Spanning Tree Protocol Concepts						
8	4.1 Explain common problems in a redundant, Layer 2 switched network 4.2 Explain how STP operates in a simple switched network. 4.3 Explain RSTP operations.	• Explain common problems in a redundant, Layer 2 switched network • Explain how STP operates in a simple switched network. • Explain RSTP operations	Multimedia Projector, Routers and Switches, Packet Tracer or eNSP or GNS3, Laptop Internet Books, Journals, Marker board, Marker, Textbooks, TeraTerm, Putty	• Investigate STP Loop Prevention. • Build a simple three-switch network with STP. • View STP operation • Disable STP and view operation again • Configure RSTP and/or Rapid per - VLAN Spanning tree (PVST+)	Guide students to; • Investigate STP Loop Prevention. • Build a simple three-switch network with STP. • View STP operation • Disable STP and view operation again • Configure RSTP and/or Rapid per - VLAN Spanning tree (PVST+)	Explain common problems in a redundant, Layer 2 switched network
GENERAL OBJECTIVE: 5.0 Know link Aggregation Technology.						
9	5.1 Explain link aggregation technology. 5.2 Explain Setup of link aggregation. 5.3 Explain troubleshooting in link aggregation	• Explain link aggregation technology. • Explain Setup of link aggregation. • Explain troubleshooting in link aggregation	Multimedia Projector, Routers and Switches, Packet Tracer or eNSP or GNS3, network cables, network toolbox, Laptop Internet Books, Journals, Marker board, Marker, Textbooks, TeraTerm, Putty	• Configure link aggregation. • Troubleshoot link aggregation. • Verify link aggregation between the switches	Guide students to; • Configure link aggregation. • Troubleshoot link aggregation. • Verify link aggregation between the switches	Explain link aggregation technology

GENERAL OBJECTIVE: 6.0 Understand Dynamic Host Configuration Protocol for IPv4/ IPv6						
10 -11	6.1 Explain DHCPv4 operations in a small- to medium-sized business network. 6.2 Explain how an IPv6 host can acquire its IPv6 configuration. 6.3 Explain the operation of SLAAC 6.4 Explain the operation of DHCPv6	• Explain DHCPv4 operations in a small- to medium-sized business network. • Explain how an IPv6 host can acquire its IPv6 configuration. • Explain the operation of SLAAC • Explain the operation of DHCPv6	Multimedia Projector, Routers and Switches, Packet Tracer or eNSP or GNS3, network cables, network toolbox, lab guides, Laptop Internet Books, Journals, Marker board, Marker, Textbooks, TeraTerm, Putty	• Configure DHCPv4: • Configure a Router as a DHCP Server • Configure DHCP Relay • Configure a Router as a DHCP Client • Verify DHCP and Connectivity. • Implement DHCPv4 • Configure DHCPv6. • Verify SLAAC address assignment on a router • Configure a Stateless DHCPv6 Server on a router • Configure a Stateful DHCPv6 Server on a router • Configure a	Guide students to; • Configure DHCPv4: • Configure a Router as a DHCP Server • Configure DHCP Relay • Configure a Router as a DHCP Client • Verify DHCP and Connectivity. • Implement DHCPv4 • Configure DHCPv6. • Verify SLAAC address assignment on a router • Configure a Stateless DHCPv6 Server on a router • Configure a Stateful DHCPv6 Server on a router • Configure a	Explain DHCPv4 operations in a small- to medium-sized business network.

				DHCPv6 Relay on a router	DHCPv6 Relay on a router	
GENERAL OBJECTIVE: 7.0 Understand Stateless Address Auto Configuration (SLAAC) and Dynamic Host Configuration Protocol for IPv6						
12	7.1 Explain how an IPv6 host can acquire its IPv6 configuration. 7.2 Explain the operation of SLAAC 7.3 Explain the operation of DHCPv6 7.4 Explain how to configure a stateful and stateless DHCPv6 server.	• Explain how an IPv6 host can acquire its IPv6 configuration. • Explain the operation of SLAAC • Explain the operation of DHCPv6 • Configure a stateful and stateless DHCPv6 server.	Multimedia Projector, Routers and Switches, Packet Tracer Laptop Internet PC/Laptop, Multimedia projector, Marker, Marker board, Practical Manual, Books	• Configure DHCPv6. • Build the Network and Configure Basic Device Settings • Verify SLAAC address assignment on a router • Configure and verify a Stateless DHCPv6 Server on a router • Configure and verify a Stateful DHCPv6 Server on a router • Configure and verify a DHCPv6 Relay on a router	Guide students to: • Configure DHCPv6. • Build the Network and Configure Basic Device Settings • Verify SLAAC address assignment on a router • Configure and verify a Stateless DHCPv6 Server on a router • Configure and verify a Stateful DHCPv6 Server on a router • Configure and verify a DHCPv6 Relay on a router	Explain how an IPv6 host can acquire its IPv6 configuration. Explain the operation of SLAAC

CLOUD COMPUTING 1

PROGRAMME: HIGHER NATIONAL DIPLOMA (HND) NETWORKING AND CLOUD COMPUTING			
COURSE: CLOUD COMPUTING I	CODE: NCC 322	Credit Unit: 3	CONTACT HOURS: 3 HOURS/WEEK
GOAL: This course is designed to equip students with theoretical and practical skills required to gain a foundational understanding of cloud computing.			
YEAR: 1 SEMESTER: 1	PRE-REQUISITE:	THEORETICAL: 2 HOURS/WEEK PRACTICAL: 1 HOURS/WEEK	
GENERAL OBJECTIVES: On completion of the course, the student should be able to:			
2.0 Understand Cloud Computing Overview.			
2.0 Understand Cloud Economics And Billing			
4.0 Understand Global Cloud Infrastructure			
5.0 Understand The Fundamentals Of Cloud Computing Security			
6.0 Understand Cloud Networking And Content Delivery			

PROGRAMME: HIGHER NATIONAL DIPLOMA (HND) NETWORKING AND CLOUD COMPUTING						
COURSE: CLOUD COMPUTING I			CODE: NCC 312		Contact Hours: 3 Hours/Week	
CREDIT UNIT: 3					Theoretical: 2 Hours/Week Practical: 1 Hour/Week	
GOAL: This course is designed to equip students with theoretical and practical skills required on cloud computing.						
Course Specification: THEORETICAL CONTENT				PRACTICAL CONTENT		
GENERAL OBJECTIVE 1.0: Understand cloud computing overview						
Week	Specific Learning Outcome	Teachers’ Activities	Learning Resources	Specific Learning Outcome	Teachers’ Activities	Evaluation
1-3	1.1 Define the term cloud computing. 					

	advantages of cloud computing 1.5 Define web services and give examples of web service providers. 1.6 Identify simple solution example and how users interact with the cloud services. 1.7 Explain the cloud adoption framework (CAF)	advantages of cloud computing • Explain web services such as networking, database, storage, application integration, etc and give examples like AWS, Microsoft Azure, Google Cloud, etc • Explain different ways to interact with cloud service(s) such as GUI management console, CLI, etc • Explain the cloud adoption framework (CAF) and state the difference between AWS CAF, Google CAF, and Microsoft Azure CAF		Demonstrate cloud adoption framework (CAF)	Guide students to demonstrate cloud adoption framework (CAF)	
GENERAL OBJECTIVE 2.0: Understand Cloud Economics and Billing						
4-6	2.1 Explain the cloud service pricing philosophy	• Explain the cloud service pricing types such as pay-as-you-go, custom pricing, free, etc for AWS,	Online Pricing Calculator, Internet, Multimedia	• Take an existing case study, and use a pricing calculator to	• Guide students on how to read estimate and use pricing	Explain fundamental pricing types.

	2.2 Explain fundamental pricing types. 2.3 Differentiate premises versus cloud costing 2.4 Explain the elements of total cost of ownership (TCO). 2.5 Explain the components to consider for TCO preparation. 2.6 Explain how to set up an organizational structure that simplifies billing and account visibility to review cost data. 2.7 Explain the functionality in an online billing	Microsoft Azure, etc. • Explain fundamental pricing characteristics for different providers. • Explain and differentiate premises versus cloud costing. • Explain Total cost of ownership • Explain the TCO consideration. • Explain the 4 steps to setup organization on a provider's platform for efficient billing and account management, etc. • Explain cloud service organizations terminologies	Projector, Computer, Lecture notes, Tutorials, Marker, Marker board, Books/Journals, PowerPoint slides	provide an estimate. • Use billing tools such as Bills, Cost Explorer, Budgets, etc.	calculator with a case study. • Guide students on how to use the billing tools.	
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	dashboard. 2.8 Explain the limitations imposed on the providers platform for setting up organizations 2.9 Explain how to use costing and billing tools.	Explain organizations security, key features and benefits Explain how to use billing tools such as Bills, Cost Explorer, Budgets, etc				
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GENERAL OBJECTIVE 3.0: Understand Global Cloud Infrastructure

7-9	3.1 Explain the purpose of global cloud infrastructure. 3.2 Explain the components that make up a cloud global cloud infrastructure 3.3 Explain factors to consider when selecting a region. 3.4 Define data center and explain where it fits in the global cloud infrastructure.	• Explain the meaning and purpose of global cloud infrastructure. • Explain the components that make up a global cloud infrastructure • Explain the factors that are considered when selecting a region for implementing the infrastructure. • Explain the components such as global regions, zones, data centers,	Online global infrastructure view for Microsoft Azure, AWS, or any other, Internet, Multimedia Projector, Computer, Lecture notes, Tutorials, Marker, Marker board, Books/Journals, PowerPoint slides	Identify global cloud infrastructures	Show global cloud infrastructure	Explain the components that make up a cloud global cloud infrastructure
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	3.5 Explain point of presence for different global cloud infrastructure 3.6 Explain different categories of services and their cloud service providers 3.7 Explain different management consoles	etc • Explain points of presence for different global cloud infrastructure from different providers • Explain common categories of services such as compute, storage, database, networking, and outline some proprietary examples specific to different providers • Explain some popular management consoles such as Azure Portal, AWS Management Console				
GENERAL OBJECTIVE 4.0: Understand the Fundamentals of Cloud Computing Security						
10-12	4.1 Explain the shared responsibility model for cloud security 4.2 Explain the components that form both the	• Explain the shared responsibility model for cloud security using models such as AWS, Azure AD, etc. • Explain the responsibility of the customer and the	Microsoft Azure, AWS, or any other, Internet, Multimedia Projector, Computer,	Create Account on AWS, Microsoft Azure platforms and others. Create and add users to groups. Explore the users and groups,	Guide students to create account, add users, create groups, add users to groups, and test sign-in to the created accounts	Explain the shared responsibility model for cloud security

	customer and the provider responsibilities in the shared model. 4.3 Explain the cloud service characteristics and security. 4.4 Explain the shared responsibilities component by answering the scenario-based quiz. 4.5 Explain the components of cloud service access and control management. 4.6 Explain the management of control and access to resources.	provider in cloud security model. • Explain and differentiate security services manage by customer and those manage by the provider with examples • Explain the concept of shared responsibilities using scenario-based quiz • Explain the access and control management essential components such as users, group, policy, and role. And explain the components using AWS IAM, Azure AD, or any other service • Explain how the access and control is managed using user and group, roles, MFA, access conditions, directory integration	Lecture notes, Tutorials, Marker, Markerboard, Books/Journals, PowerPoint slides	Sign-in and test users account. Demonstrate how to secure data on the cloud platforms	Guide students to demonstrate how to secure data on the cloud platforms	
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	4.7 Explain account access types and how to secure a new account. 4.8 Explain how to secure data on the cloud platforms 4.9 Explain how to ensure compliance with regulatory requirements	• Explain different account access types such as root, etc and how to secure a new account • Explain how to secure data at rest or in transit using encryption provided on the cloud platform. Use AWS S3 buckets or Azure Blob Storage for elaboration • Explain how to ensure compliance with regulatory requirements on the platforms				
GENERAL OBJECTIVE 5.0: Understand Cloud Networking and Content Delivery.						
13-15	5.1 Explain the basics of networking 5.2 Explain IP address and their versions. 5.3 Explain the concept of subnetting and CIDR 5.4 Explain OSI model.	• Explain what is a network • Explain the concept of IP addressing and subnets, IPv4, IPv6, and CIDR • Explain OSI model and the possible protocols found in each layer	AWS Management Console, Azure VN Designer, or any other, Pictorial network diagrams, Internet, Multimedia Projector, Computer,	• Design a cloud virtual network diagram, • Build a virtual network and launch a web server, • Create a virtual network, create additional subnets, create virtual network security group,	• Guide the student to design a cloud virtual private network, • Guide the students to build a virtual network and launch a web server, • Guide the	Explain cloud virtual network

	5.5 Explain IP address and subnet. 5.6 Explain cloud virtual network 5.7 Explain the concept of cloud virtual network in relation to cloud computing organizational structure 5.8 Describe how IP address, routes, and route tables are used in cloud virtual networks 5.9 Describe the concept of elastic network interface 5.9 Describe the concept of cloud virtual networking	Explain IP address and subnet • Explain cloud virtual networks • Explain the concept of cloud virtual network using AWS VPC, and Azure VN, and how the technology is used in a region, zones, and subnets. • Explain how IP address, routes, and route tables are used in cloud virtual networks • Explain elastic network interface and how it can be connected to instances • Explain with help of sample designs the concept of cloud virtual networking such as internet and NAT gateway, sharing, peering, site-to-site VPN,	Lecture notes, Tutorials, Marker, Markerboard, Books/Journals, PowerPoint slides	and launch a web server instance.	students to create a virtual network, create additional subnets, create virtual network security group, and launch a web server instance.	
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	5.10 Explain the cloud virtual network security groups 5.11 Explain network access control lists (ACLs) for cloud virtual networks and compare between security groups and network ACLs 5.13 Explain the concept of cloud DNS and cloud DNS resolution 5.13 Explain how multi-region deployment and fail-over can be used to improve availability 5.14 Explain the concept of CDN	direct connect, endpoints, and transit gateway • Explain the cloud virtual network security groups • Explain network access control lists (ACLs) for cloud virtual networks and compare between security groups and network ACLs • Explain the high availability and scalability of cloud DNS using Amazon 53 and Azure DNS and how it support several routing policies • Explain how multi-region deployment and fail-over can be used to improve availability				
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		• Explain how CDN is used to accelerate delivery of content using either AWS CloudFront or Azure CDN				
ASSESSMENT: The continuous assessment; tests, quizzes will be awarded 40% of the total score. The end of the Semester Examination will make up for the remaining 60% of the total score.						

ADVANCED STATISTICS FOR COMPUTING

PROGRAMME: HIGHER NATIONAL DIPLOMA (HND) NETWORKING AND CLOUD COMPUTING		
Course Title: Advanced Statistics for Computing	Course Code: NCC 323	Contact Hours: 3
Year: 1 Semester: 1		Theoretical: 1hour /week Practical: 2 hour /week
Goal: This course is designed to enable students to acquire basic knowledge of SPSS Package Tools		
General Objectives: On completion of this course the diplomate, should be able to: 1.0 Understand the Main Features Of SPSS (Statistical Package for Social Science) 2.0 Understand Variable and Data Input Transformation in SPSS 3.0 Understand How to Perform Descriptive Analyses with SPSS and Ms Excel ER 4.0 Know How to Perform Common Parametric and Non-Parametric Test 5.0 Understand How to Perform Simple Regression and Multivariate Analyses		

PROGRAMME: HIGHER NATIONAL DIPLOMA (HND) NETWORKING AND CLOUD COMPUTING						
Course Title: Advanced Statistics for Computing			Course Code: NCC 323		Contact Hours: 3	
CREDIT UNITS: 3				Theoretical: 1 hour /week Practical: 2 hour /week		
Theoretical Content				Practical Content		
General Objective 1: Understand the main features of SPSS (Statistical Package for Social Science)						
Week	Specific Learning Outcomes	Teacher’s activities	Resources	Specific Learning Outcomes	Teacher’s activities	Evaluation
1-2	1.1 Define SPSS Package 1.2 Explain SPSS general features 1.3 Explain the Importance of SPSS 1.4 Describe File management in SPSS 1.5 Explain data file Storage and Retrieval	• Explain the main features of SPSS • Explain the general aspect, workflow and critical issues • Explain Functions, Menus and commands • Explain file management in SPSS • Explain data file Storage and Retrieval	Books of recorded statistics Internet	Install SPSS Identify the features in SPSS	Guide students to: • Install SPSS • Identify the features in SPSS	Explain SPSS Identify general Features of SPSS Explain Sorting, Transpose in SPSS Explain how to store and retrievefiles

		Explain the importance of SPSS				
General Objective 2.0 Understand Variable and Data Input Transformation in SPSS						
3-4	2.1 Define Variable, 2.2 Describe Manual Data input 2.3 Explain Automated Data Input and file import 2.4 Explain Data Transformation 2.5 Explain syntax files and scripts 2.6 Explain Output Management	- Define Variable and Explain variable view spreadsheet - Explain Manual Data Entry - Explain Data Transformation - Explain Syntax files and script - Describe how to generate data and Import file using computer system	Textbooks Lecture notes Internet PCs SPSS Textbooks PCs SPSS	Demonstrate the concept of Variable Use computer system to generate data	Demonstrate the concept of Variable Illustrate how to generate data online Illustrate how to Transform Data	Describe variable Explain the various methods of data Input Explain Data Transformation

General Objective 3: Understand how to perform descriptive analyses with SPSS						
5-6	3.1 Explain Frequencies	Explain Frequencies	Textbooks Ms. Excel	Categories various data	Explain and supervise student exercises and student work	Explain frequency,
	3.2 Explain Descriptive Analysis	Explain Descriptive analysis	Textbooks Ms. Excel Hard disk, Flash drive, CD, internet SPSS etc	collected Illustrate Crosstab and Chart	Explain crosstab and Chart	Explain Crosstab and Chart
	3.3 Explain Crosstab	Explain Crosstab				
	3.4 Explain Charts	Explain and discuss Charts				
General Objective 4 : Know how to perform common parametric and non-parametric test						
7-8	4.1 Identify and Explain different statistical test: Mean, T test, One-way ANOVA, Non Parametric test, Normality test	Explain and discuss various types of statistical tests	Textbooks Statistical tables	Identify the various types of statistical table	Demonstrate how to identify the various types of statistical tables	Enumerate the various types of statistical tables
	4.3 Explain Correlation and Regression: - Linear Correlation and Regression, - Multiple regression (Linear)	Explain Linear Correlation and Regression Explain Multiple regression (Linear)	Statistical tables, PCs, Charts, Ms. Excel SPSS	Demonstrate how to construct scattered diagrams, frequency tables and graphs	Demonstrate by examples how to construct scattered diagrams, frequency tables and graphs	Explain how to construct frequency tables and graphs Enumerate the merits and demerits of charts and diagrams

	General Objective 5: Understand how to perform advanced regression and multivariate analyses					
9	5.1 Explain Factor Analysis	Define and Explain Factor Analysis	Text books SPSS PCs SPSS Package	Analyze data using Factor analysis Analyze data using cluster analysis Analyze data using SPSS	Guide students to analyze data using factor analysis Guide students to analyze data using SPSS	Explain factor and cluster analysis Use SPSS to analyze data
	5.2 Explain Cluster Analysis	Define and Explain Cluster Analysis				
10-12	5.3 Analyze Data using SPSS	Explain how to analyze data using SPSS				

ADVANCE WIRELESS NETWORKS

PROGRAMME: HIGHER NATIONAL DIPLOMA (HND) NETWORKING AND CLOUD COMPUTING		
COURSE: ADVANCE WIRELESS NETWORKS	COURSE CODE: NCC 325	Contact Hours: 3 Hours/Week
	CREDIT UNIT: 3	Theoretical: 2 Hours/Week
Year: 1 Semester: 1	Pre-requisite:	Practical: 1 Hour/Week
GOAL: This course is designed to equip students with comprehensive understanding of WLAN Technologies and its Deployment.		
GENERAL OBJECTIVES: On Completion Of The Course, The Student Should Be Able To: 1.0 Understand WLAN Technology, Products and VRP 2.0 Understand WLAN Fundamentals and Configuration 3.0 Understand the WLAN Security, Operation and Management 4.0 Understand Historical Progression and Advancements in Modern Wireless Communication Systems. 5.0 Understand 5G Industrial Applications and Solutions		

PROGRAMME: HIGHER NATIONAL DIPLOMA (HND) NETWORKING AND CLOUD COMPUTING						
COURSE: ADVANCE WIRELESS NETWORKS		COURSE CODE: NCC 325			Contact Hours: 3 Hours/Week	
		CREDIT UNIT: 3			Theoretical: 2 Hours/Week	
Year: ONE 1 Semester: 11		Pre-requisite:			Practical: 1 Hour/Week	
COURSE SPECIFICATION: THEORETICAL AND PRACTICAL						
GOAL: This course is designed to equip students with comprehensive understanding of WLAN Technologies and its Deployment.						
GENERAL OBJECTIVE 1.0: Understand WLAN technology and products						
THEORETICAL CONTENT				PRACTICAL CONTENT		
Week	Specific Learning Outcome	Teacher’s Activities	Resources	Specific Learning Outcome	Teacher’s Activities	Evaluation
1 – 3	1.1 Explain WLAN technical principles and products. 1.2 Explain the basic wireless configuration methods 1.3 Explain how to upgrade WLAN devices.	• Explain WLAN technical principles and products • Explain the basic wireless configuration method. • Explain how to upgrade WLAN devices	Text books, multi-media Projector, Lecture notes, Lab guide, simulation tools, Switches, Routers, Access Point s (APs), Access Controllers (ACs)	• Perform basic wireless LAN configuration • Perform WLAN Device Upgrade (ACs and APs)	• Guide students on basic wireless LAN configuration • Guide the students on how to Upgrade WLAN Devices	Explain how to upgrade WLAN devices.
GENERAL OBJECTIVE: 2.0 Understand WLAN Fundamentals and Configuration						
4 – 6	2.1 Explain the working principles of Layer 2 WLAN networking and Layer 3 WLAN networking. 2.2 Explain the various WLAN service configuration process and commands. 2.3	• Describe Layer 2 WLAN networking and Layer 3 WLAN networking. • Explain the WLAN service configuration process and commands.	Text books, multi-media Projector, Lecture notes, Lab guide, simulation tools, Switches, Routers, Access	• Design and configure Layer 2 Off-Path WLAN Networking • Design and configure Layer 3 Off-Path WLAN Networking Experiment (CLI) • Configure Layer 3	• Guide the students to design and configure Layer 2 Off-Path WLAN Networking • Guide the students to design and configure Layer 3 Off-Path WLAN Networking	Explain the various WLAN service configuration process and commands.

	2.4 Explain WLAN radio resource management.	Explain WLAN radio resource management.	Point s (APs), Access Controllers (ACs)	Off-Path WLAN Networking Experiment (Web) Configure WLAN Radio Resource Management	Experiment (CLI) - Demonstrate Layer 3 Off-Path WLAN Networking Experiment (Web) - Demonstrate WLAN Radio Resource Management	
GENERAL OBJECTIVE: 3.0 Understand WLAN Security, Operations and Management						
7 - 8	3.1 Define various WLAN security technologies . 3.2 Explain how to perform basic troubleshooting WLAN methods. 3.3 Explain the WLAN deployment solution	- Explain WLAN security technologies. - Explain the basic WLAN troubleshooting methods. - Explain the WLAN deployment solution	Text books, multi-media Projector, Lecture notes, Lab guide, simulation tools, Switches, Routers, Access Point s (APs), Access Controllers (ACs)	Perform some basic WLAN Troubleshooting.	Demonstrate some basic WLAN Troubleshooting	Define various WLAN security technologies
GENERAL OBJECTIVE: 4.0 Understand the Historical Progression and Advancements In Modern Wireless Communication Systems.						
9 – 11	4.1 Explain the evolution of wireless communication systems, highlighting key milestones. 4.2 List the characteristics and benefits of Wi-Fi 6 (802.11ax) technology. 4.3	- Describe the evolution of wireless communication systems, highlighting key milestones. - Explain the various characteristics and benefits of Wi-Fi 6 (802.11ax)	Text books, multi-media Projector, Lecture notes, Computers, Animation videos/illustrations.			Explain the evolution of wireless communication systems, highlighting key milestones.

	4.4 Outline the features, use cases, and advancements brought by 5G networks. 4.4 Describe 5G Network architecture and key technologies 4.5 Explain 5G industry chain progress 4.1 4.2 Identify key network performance and objectives in three 5G service scenarios 4.3 Explain 5G spectrum resources and country-specific distribution and allocation 4.4 Outline key technologies of 5G new radio: uplink and downlink decoupling, new coding technologies, F-OFDM, etc.	technology. • Explain the features, use cases, and advancements brought by 5G networks. • Explain 5G Network architecture and key technologies • Explain 5G industry chain progress 4.5 Explain key network performance and objectives in three 5G service scenarios • Explain 5G spectrum resources and country-specific distribution and allocation • Explain key technologies of 5G new radio: uplink and downlink decoupling, new coding technologies, F-OFDM, etc.				
GENERAL OBJECTIVE: 5.0 Understand 5G Industrial Applications and Solutions						
12 -14	5.1 List IoV Technical Features, and Industry and Standards Progress	• Explain IoV Technical Features, and Industry and Standards Progress	Text books, multi-media Projector, Lecture notes,			List IoV Technical Features, and Industry and

	5.2 Outline 5G IoV Application Scenarios and Solutions 5.3 Outline 5G Smart Healthcare Application Scenarios and Solutions 5.4 Outline 5G Smart EducationsApplication Scenarios and Technical Requirements 5.5 Outline 5G Smart Grid Application Scenarios and Technical Requirements 5.6 Outline 5G Smart Port Application Scenarios and Technical Requirements 5.7 Outline 5G Smart Campus Application Scenarios and Technical Requirements 5.8 Outline 5G Smart Manufacturing Application Scenarios and Technical Requirements	• Explain 5G IoV Application Scenarios and Solutions • Explain 5G Smart Healthcare Application Scenarios and Solutions • Explain 5G Smart Education Application Scenarios and Technical Requirements • Explain 5G Smart Grid Application Scenarios and Technical Requirements • Explain 5G Smart Port Application Scenarios and Technical Requirements • Explain 5G Smart Campus Application Scenarios and Technical Requirements • Explain 5G Smart Manufacturing Application Scenarios and Technical Requirements	Computers, Animation videos/illustrations.			Standards Progress
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YEAR TWO SEMESTER ONE COURSES

RESEARCH METHODOLOGY IN NETWORKING AND CLOUD COMPUTING

PROGRAMME: HIGHER NATIONAL DIPLOMA NETWORKING AND CLOUD COMPUTING			
Course: RESEARCH METHODOLOGY IN NETWORKING AND CLOUD COMPUTING	Course Code: NCC 411	Credit Unit: 2	Contact Hours: 60 (2 hrs lecture, 2 hrs practical)
Year: 11 Semester: 1	Pre requisite		
GOAL: This course is designed to provide students with a comprehensive understanding of the fundamental research principles, methodologies, and best practices essential for conducting effective and innovative research in Networking and Cloud Computing			
General Objectives: 1.0 Understand NCC Research scope. 2.0 Evaluate Research Literature 3.0 Design Robust NCC Experiments 4.0 Apply Ethical Research Practices 5.0 Utilize NCC Tools and Frameworks 6.0 Collaborate in Research Projects			

PROGRAMME: HIGHER NATIONAL DIPLOMA NETWORKING AND CLOUD COMPUTING							
Course: RESEARCH METHODOLOGY IN NETWORKING AND CLOUD COMPUTING			Course Code: NCC 411		Credit Unit: 2	Contact Hours: 60 (2 hrs lecture, 2 hrs practical)	
Year: II Semester: I							
GOAL: This course is designed to provide students with a comprehensive understanding of the fundamental research principles, methodologies, and best practices essential for conducting effective and innovative research in Networking and Cloud Computing							
Course Specification: THEORETICAL CONTENT					PRACTICAL CONTENT		
General /Objective 1.0: Understand NCC Research scope.							
Week	Specific Learning Outcome	Teachers’ Activities	Learning Resources	Specific Learning Outcome	Teachers’ Activities	Evaluation	
1-2	1.1 Explain the fundamental concepts of NCC and its applications.	Discuss NCC concepts and real-world applications.	Reading materials, case studies, and research papers for students to explore current NCC research trends.	Analyze real-world NCC projects or case studies and identify their key components and challenges.	Guide student to analyze real-world NCC projects or case studies and identify their key components and challenges.	Explain the fundamental concepts of NCC and its applications. Enumerate key milestones and historical developments in NCC research.	
	1.2 Explain key milestones and historical developments in NCC research.	Discuss timeline of significant events in NCC research history and facilitate discussions		Develop a simple NCC prototype or model using a programming language or NCC toolkit.	Guide student to Develop a simple NCC prototype or model using a programming language or NCC toolkit.	Explain the current state of NCC research, including major trends and breakthroughs.	
	1.3 Explain the current state of NCC research, including major trends and breakthroughs.	Discuss the current state of NCC research, including major trends and breakthroughs					
	1.4 Explain ethical considerations and challenges related to NCC research.	Discuss on ethical issues in NCC research.					

	1.5 Explain the interdisciplinary nature of NCC and its collaboration with other fields	Discuss the interdisciplinary nature of NCC and its collaboration with other fields				
General Objective 2.0: Evaluate Research Literature						
3-4	2.1 Explain key components of research literature.	Discuss key components of research literature, such as abstracts, introductions, methodologies, results, and conclusions.	Reading materials, case studies, and research papers for students to explore current NCC research trends.	Evaluate a research paper and present findings in a structured manner.	Guide student to evaluate a research paper and present findings in a structured manner.	Define key components of research literature.
	2.2 Explain the credibility and reliability of research sources and identify potential biases.	Discuss the credibility and reliability of research sources and identify potential biases.		Compare and contrast research methodologies used in two different papers.	Guide student to Compare and contrast research methodologies used in two different papers.	Explain the credibility and reliability of research sources and identify potential biases.
	2.3 Explain the criteria to evaluate the quality and relevance of research articles.	Discuss the appropriate criteria to evaluate the quality and relevance of research articles.		Prepare a literature review by synthesizing information from multiple research articles.	Guide student to Prepare a literature review by synthesizing information from multiple research articles.	
	2.4 Explain the strengths and weaknesses of different research methodologies used in the literature.	Discuss the strengths and weaknesses of different research methodologies used in the literature.				
	2.5 Explain how to synthesize information from multiple research papers to form a	Discuss how to Synthesize information from multiple research papers to form a				

	coherent and well-supported argument.	coherent and well-supported argument.				
General Objective 3.0: Design Robust NCC Experiments						
5-6	3.1 Explain the key components of a robust NCC experiment. 3.2 Explain how to formulate research questions suitable for NCC experimentation. 3.3 Explain how to Select appropriate datasets for NCC experiments. 3.4 Explain how to apply relevant machine learning algorithms to solve specific problems. 3.5 Explain how to analyze and interpret the results of NCC experiments. f. Implement proper techniques for model evaluation and validation. 3.6 Explain ethical considerations in NCC experiment design.	Discuss the key components of a robust NCC experiment. Discuss how to formulate research questions suitable for NCC experimentation. Discuss how to Select appropriate datasets for NCC experiments. Discuss how to apply relevant machine learning algorithms to solve specific problems. Discuss how to analyze and interpret the results of NCC experiments. Discuss ethical considerations in NCC experiment design.	Marker board, Markers, Documentary films, Multimedia Projector and Screen, Case studies, Power point slides, Internet, Textbooks, Computer, Flip charts, Policy documents, research papers, etc.	Apply relevant machine learning algorithms to solve specific problems. Implement proper techniques for model evaluation and validation. Demonstrate ethical considerations in NCC experiment design.	Guide student to apply relevant machine learning algorithms to solve specific problems. Guide student to Implement proper techniques for model evaluation and validation. Demonstrate to student ethical considerations in NCC experiment design.	Explain the key components of a robust NCC experiment. Formulate research questions suitable for NCC experimentation. Explain how to Select appropriate datasets for NCC experiments.

General Objective 4.0: Apply Ethical Research Practices:						
7-8	4.1 Explain the principles of ethical research and their importance in academic and professional settings.	Discuss ethical research principles, codes, and guidelines.	Marker board, Markers, Documentary films, Multimedia Projector and Screen, Case studies, Power point slides, Internet, Textbooks, Computer, Flip charts, Policy documents, research papers, etc.	Develop a research proposal that incorporates ethical considerations.	Guide student to develop a research proposal that incorporates ethical considerations.	Describe the principles of ethical research and their importance in academic and professional settings.
	4.2. Explain potential ethical issues and challenges that may arise during the research process.	Discuss potential ethical issues and challenges that may arise during the research process.		Conduct practical on ethical review of an existing research study. Prepare an ethical data management plan for a research project.	Guide student to conduct practical on ethical review of an existing research study.	Explain potential ethical issues and challenges that may arise during the research process.
	4.3 Explain the ethical guidelines and protocols to design, conduct, and report research studies.	Discuss the ethical guidelines and protocols to design, conduct, and report research studies.			Guide student to prepare an ethical data management plan for a research project.	Describe ethical guidelines and protocols to design, conduct, and report research studies.
	4.4. Explain the ethical implications of research findings and propose appropriate solutions.	Discuss the ethical implications of research findings and propose appropriate solutions.				
	4.5 Explain awareness of the ethical use of data, including privacy, confidentiality, and informed consent.	Discuss awareness of the ethical use of data, including privacy, confidentiality, and informed consent.				
General Objective 5.0: Utilize NCC Tools and Frameworks						
9-10	5.1. Explain proficiency in using popular NCC tools and frameworks	Discuss proficiency in using popular NCC tools and frameworks, such as Mindspore, TensorFlow or	Marker board, Markers, Documentary films, Multimedia Projector	Build and train an image classification model using TensorFlow.	Guide student to Build and train an image classification model using	Explain NCC tools to preprocess data and prepare it for model training and evaluation.

	5.2. Explain AI tools to preprocess data and prepare it for model training and evaluation. 5.3. Explain and train NCC models using the selected frameworks to solve specific problems 5.4. Explain the performance of NCC models and fine-tune hyper parameters for optimization. 5.5. Explain how to integrate NCC models into real-world applications and interpret their outputs.	PyTorch. Discuss exercises for data preprocessing tasks using NCC tools. Discuss coding sessions and projects where students build NCC models using selected frameworks. Discuss case studies on model evaluation and hyper parameter tuning. Discuss NCC model integration into applications.	and Screen, Case studies, Power point slides, Internet, Textbooks, Computer, Flip charts, Policy documents, research papers, etc. Mindspore, TensorFlow or PyTorch	Fine-tune a pre-trained natural language processing model with PyTorch. Implement a simple NCC-based recommendation system using a relevant framework.	TensorFlow. Guide student to fine-tune a pre-trained natural language processing model with PyTorch. Guide student to Implement a simple NCC-based recommendation system using a relevant framework.	Explain and train NCC models using the selected frameworks to solve specific problems
General Objective 6.0: Collaborate in Research Projects						
11-13	6.1 Explain effective communication and teamwork skills in a research project setting. 6.2 Explain actively to research discussions and propose valuable ideas and insights. 6.3 Explain the management of project	Discuss effective communication and teamwork skills in a research project setting. Discuss actively to research discussions and propose valuable ideas and insights. Discuss with peers to divide tasks, set goals,	Marker board, Markers, Documentary films, Multimedia Projector and Screen, Case studies, Power point slides, Internet, Textbooks, Computer, Flip charts, Policy Mindspore, TensorFlow or	Collaborate in a research project with a team of peers. Conduct data analysis and interpretation as part of the research project	Assign students to research project teams based on their interests and expertise. Provide guidance and support to student during data analysis and	Explain actively to research discussions and propose valuable ideas and insights. Discuss with peers to divide tasks, set goals, and manage project timelines.

	timelines. 6.4 Explain in data collection, analysis, and interpretation in a research project. 6.5 Explain research findings coherently and professionally to an audience.	and manage project timelines. Discuss data collection, analysis, and interpretation in a research project. Discuss research findings coherently and professionally to an audience.	PyTorch documents, research papers, etc.	Deliver a group presentation on the research project's findings	interpretation stages. Guide students to present research project's findings	
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PROJECT MANAGEMENT IN NETWORKING AND CLOUD COMPUTING

PROGRAMME: HIGHER NATIONAL DIPLOMA IN NETWORKING AND CLOUD COMPUTING		
COURSE TITLE: PROJECT MANAGEMENT IN NETWORKING AND CLOUD COMPUTING	Course Code: NCC 412	CONTACT HOURS: 2 hours/week
YEAR: II SEMESTER I	CREDIT UNITS: 2	Theoretical: 1 hours/week
	PRE-REQUISITE:	Practical: 1 hours/week
GOAL: This course is designed to equip students with the knowledge and skills necessary to manage networks and Cloud projects.		
General Objectives: On completion of the course, the student should be able to: 1.0 Understand Project management tools and software relevant to Networking and Cloud Computing projects. 2.0 Understand the fundamentals of project management and its application to Cloud Computing and Networking projects. 3.0 Appreciate the unique challenges and requirements of managing Networks and Cloud Projects 4.0 Know how to develop project plan for Networking and Cloud Computing. 5.0 Know how to apply best practices for team management in Networking and Cloud Computing projects. 6.0 Understand ethical considerations, privacy issues, and legal implications inherent in Networking and Cloud Computing project management.		

PROGRAMME: HIGHER NATIONAL DIPLOMA IN NETWORKING AND CLOUD COMPUTING						
Course: PROJECT MANAGEMENT IN NETWORKING AND CLOUD COMPUTING			COURSE CODE: NCC 412		CONTACT HOURS: 2 hours/week	
CREDIT UNITS: 2			Theoretical: 1 hour/week		Practical: 1 hour/week	
GOAL: This course is designed to equip students with the knowledge and skills necessary to manage Networks and Cloud Computing projects.						
General Objective: 1.0 Understand project management tools and software relevant to Networking and Cloud Computing projects						
Week	Specific Learning Outcomes	Teachers Activities	Learning Resources	Specific Learning Outcomes	Teachers Activities	Evaluation
1-2	1.1 Explain the range of project management tools and software and their application in managing projects. 1.2 Explain various project management tools, such as Jira, Trello, and Asana, among others. 1.3 Explain the appropriate tools based on the unique requirements for different Network and Cloud Computing projects. 1.4 Explain the role of these tools in team collaboration, project tracking, resource management, and risk management in Network and Cloud Computing	- Explain various project management tools through demonstrations or tutorials. - Explain the appropriate tools based on the unique requirements of different Network and Cloud Computing projects. - Explain the role of these tools in team of collaboration, project tracking, resource	Computer lab or personal computers with internet access. Subscription or access to various project management tools (e.g., Jira, Trello, Asana). Training materials or guides for using these tools.	- Simulate Network project using these tools for project management. - Simulate Cloud Computing project using these tools for project management.	Design activities where students identify, analyze, and mitigate biases in given datasets	Explain the appropriate tools based on the unique requirements of different Network and Cloud Computing projects.

	projects.	management, and risk management in Network and Cloud Computing projects.				
General Objective 2.0: Understand the fundamentals of project management and its application to Cloud computing and Networking projects.						
1-4	2.1 Explain the Principles of Project Management 2.2 Explain Project Initiation, Planning, Execution, Control, And Closure. 2.3 Explain Key Differences Between Managing Networks and Cloud Projects and Other Types of Projects. 2.4 Explain The Specific Challenges of Networks and Cloud Projects	• Explain The Fundamentals of Project Management, Focusing On Its Application In Networks and Cloud Projects. • Explain Key Differences Between Managing Networking and Cloud Projects And Other Types Of Projects. • Discuss The Specific Challenges Of Managing Networks Projects.	Learning Management System For Content Delivery. Access To Case Studies Related to Networking and Cloud Project Management. Access To Project Management Software or Tools.	• Use Project Management Software and Tools To Execute Cloud Project (Case Study 1) • Use Project Management Software and Tools to Execute Network Project (Case Study 1)	• Guide Students to use Project Management Software and Tools	Explain The Principles Of Project Management Including Project Initiation, Planning, Execution, Control, And Closure.
General Objective 3.0 Appreciate the unique challenges and requirements of managing Networking and Cloud projects						
5-6	3.1 Explain the unique challenges involved in Networking and Cloud projects.	• Explain the unique challenges involved in	Learning Management System for content delivery.	• Use Project management software and tools to execute Cloud	• Guide students to use Project Management	Explain the unique challenges involved in

	3.2 Explain strategies to mitigate challenges in Networking and Cloud projects	Networks and Cloud projects. Explain strategies to mitigate challenges in Network and Cloud projects.	Access to case studies related to Networking project management. Microsoft Project	project (Case Study 2) Use Project management software and tools to execute Networking project (Case Study 2)	software and tools	Networking projects
General Objectives: 4.0 Know how to develop a project plan for Networking and Cloud computing						
7-9	4.1 Explain the scope of Networking and Cloud projects 4.2 Explain how to develop a feasible timeline for Networking and Cloud projects 4.3 Explain accounting for various stages of the development process. 4.4 Describe resource planning and allocation for Networking and Cloud projects,	Explain various components of a comprehensive project plan for Networking project. Explain various components of a comprehensive project plan for Cloud Computing project. Explain how to set realistic goals and objectives Describe resource planning and allocation for Networking and Cloud Computing	Learning Management System (Blackboard, Google Classroom) Project management software (Microsoft project, Primavera)	Use Project management software and tools to execute Cloud project (Case Study 3) Use Project management software and tools to execute Networking project (Case Study 3)	Guide students to use Project Management software and tools	Explain risk management within the context of Networking projects, including identifying, analyzing, and mitigating potential risks. Explain risk management within the context of Cloud Computing projects, including identifying, analyzing, and mitigating

	recognizing the unique resource needs of Networking and Cloud projects. 4.5 Explain risk management within the context of Networking and Cloud projects, including identifying, analyzing, and mitigating potential risks.	projects recognizing the unique resource needs of the projects. Explain risk management within the context of Network and Cloud projects, including identifying, analyzing, and mitigating potential risks.				potential risks.
General Objective: 5.0 Know how to apply best practices for team management in Networking and Cloud projects						
10-12	5.1 Explain the challenges of managing cross-functional teams in Networking and Cloud Computing projects. 5.2 Explain strategies for effective communication and collaboration within a cross-functional Networking and Cloud Computing projects.	• Explain the best practices of team management, particularly in the context of Networking and Cloud computing projects. • Explain strategies for effective communication and collaboration within a cross-functional Networking team.	Learning Management System for content delivery. Access to case studies, literature, and industry reports on team management in Network or Cloud Computing projects. Project management software and collaboration tools that students can	• Use Project management software and tools to execute Cloud project (Case Study 4) • Use Project management software and tools to execute Network project (Case Study 4)	• Guide students to use Project Management software and tools	Explain the nuances of managing remote teams, particularly in the context of Networking and Cloud computing projects.

	5.3 Explain how to leverage project management tools and technology for effective team management in Network and Cloud Computing projects.	Explain the nuances of managing remote teams, particularly in the context of Networking and Cloud computing projects.	use to practice managing virtual teams.			
General Objective 6.0 Understand the ethical considerations, privacy issues, and legal implications inherent in Networking and Cloud project management						
	6.1 Explain the ethical considerations in managing Networking and Cloud Computing projects, including issues of fairness, accountability, and transparency.	Discuss the ethical, privacy, and legal issues in Networking and Cloud Computing project management.	Classroom or virtual learning platform for lectures and discussions. Access to research papers, articles, and case studies related to ethical, privacy, and legal issues related to Networking and Cloud Computing.	Use Project management software and tools to execute Cloud project (Case Study 5)	Guide students to use Project Management software and tools	Describe privacy issues in networking and cloud computing projects, especially related to data management.
	6.2 Explain privacy issues in Networking and Cloud Computing projects.	Illustrate how these issues can arise in real-world Networking projects and how they can be managed.		Use Project management software and tools to execute Networking and cloud projects (Case Study 5)		
	6.3 Explain the legal implications of Networking and Cloud Computing projects, such as intellectual property rights, compliance	Explain the potential implications of ignoring above considerations and issues				

	with all established laws, and other regulations.					
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ROUTING AND SWITCHING II

PROGRAMME: HIGHER NATIONAL DIPLOMA NETWORKING AND CLOUD COMPUTING		
COURSE: ROUTING AND SWITCHING II	COURSE CODE: NCC 413	Contact Hours: 4 hours /week
	CREDIT UNITS: 4	Theoretical: 1 hours /week
Year: 11 Semester: 1	Pre-requisite:	Practical: 3 hours /week
GOAL: This Course is designed to equip students with advanced skills in Routing and Switching Technologies.		
GENERAL OBJECTIVES: On completion of this course, the students should be able to; 1.0 Understand First Hop Redundancy Protocols 2.0 Understand LAN Security Concepts 3.0 Understand Switch Security Configuration 4.0 Understand WLAN configuration 5.0 Know Routing protocols 6.0 Understand IP Static Routing 7.0 Understand Troubleshooting Static and Default Routes		

PROGRAMME: HIGHER NATIONAL DIPLOMA NETWORKING AND CLOUD COMPUTING						
COURSE: ROUTING AND SWITCHING II		COURSE CODE: NCC 413			Contact Hours: 4 hours /week	
		CREDIT UNITS: 4			Theoretical: 1 hours /week	
Year: 11 Semester: I		Pre-requisite:			Practical: 3 hours /week	
GOAL: This Course is designed to equip the students with advanced skills in Routing and Switching Technologies.						
GENERAL OBJECTIVE 1.0: Understand First Hop Redundancy Protocols						
THEORETICAL CONTENT				PRACTICAL CONTENT		
Week	Specific Learning Outcome	Teacher’s Activities	Resources	Specific Learning Outcome	Teacher’s Activities	Evaluation
1 - 2	1.1 Explain the operations of first-hop redundancy protocols.	Explain the operations of first-hop redundancy protocols.	Multimedia Projector, Cisco Routers and Switches, Packet Tracer Laptop Internet eNSP	• Configure and test VRRP and HSRP	• Guide students to configure and test VRRP and HSRP	Explain VRRP and HSRP
	1.2 Explain VRRP	• Explain VRRP				
	1.3 Explain HSRP	• Explain HSRP				
GENERAL OBJECTIVE: 2.0 Understand LAN Security Concepts						
3	2.1 Explain how to use endpoint security to mitigate attacks 2.2 Explain how AAA and 802.1x are used to authenticate LAN endpoints and devices 2.3 Explain Layer 2 vulnerabilities	• Explain how to use endpoint security to mitigate attacks • Explain how AAA and 802.1x are used to authenticate LAN endpoints and devices • Explain Layer 2 vulnerabilities	Multimedia Projector, Cisco Routers and Switches, Packet Tracer Laptop Internet eNSP			Explain how to use endpoint security to mitigate attacks

	2.4 Explain how a MAC address table attack compromise LAN security	Explain how a MAC address table attack compromise LAN security				
	2.5 Explain how LAN attacks compromise LAN security	Explain how LAN attacks compromise LAN security				
GENERAL OBJECTIVE: 3.0 Understand Switch Security Configuration						
4 - 5	3.1 Explain port security 3.2 Explain DTP and native VLAN. 3.3 Explain DHCP snooping. 3.4 Explain ARP inspection 3.5 Explain Port Fast and BPDU Guard	Explain port security Explain DTP and native VLAN Explain how to configure DHCP snooping to mitigate DHCP attacks. Explain how to configure ARP inspection to mitigate ARP attacks. Explain how to configure Port Fast and BPDU Guard to mitigate STP Attacks.	Multimedia Projector, Cisco Routers and Switches, Packet Tracer Laptop Internet Multimedia Projector, Cisco Access Point, Wireless LAN Controller (WLC) Packet Tracer, SNMP Server Laptop Internet eNSP	Configure; Switch Security and Port Security Secure unused ports Mitigate; - VLAN hopping attacks - DHCP attacks - ARP attacks - STP attacks Verify the switch security configuration	Guide students to; Configure; Switch Security and Port Security Secure unused ports Mitigate; - VLAN hopping attacks - DHCP attacks - ARP attacks - STP attacks Verify the switch security configuration	Explain DHCP snooping.

GENERAL OBJECTIVE: 4.0 Understand WLAN configuration						
6	4.1 Explain WLAN technology and standards.	Explain WLAN technology and standards.	Multimedia Projector, Cisco Routers and Switches, Packet Tracer Laptop Internet Multimedia Projector, Cisco Access Point, Wireless LAN Controller (WLC) Packet Tracer, SNMP Server Laptop Internet eNSP	Configure a Wireless Network Configure a Basic WLAN on the WLC Configure a WPA2 Enterprise WLAN on the WLC Troubleshoot WLAN Issues	Guide students to: Configure a Wireless Network Configure a Basic WLAN on the WLC Configure a WPA2 Enterprise WLAN on the WLC Troubleshoot WLAN Issues	
	4.2 Explain the components of a WLAN infrastructure.	Explain the components of a WLAN infrastructure.				
	4.3 Explain how wireless technology enables WLAN operation.	Explain how wireless technology enables WLAN operation.				
	4.4 Explain how a WLC uses CAPWAP to manage multiple APs.	Explain how a WLC uses CAPWAP to manage multiple APs.				
	4.5 Explain channel management in a WLAN.	Explain channel management in a WLAN.				
	4.6 Explain threats to WLANs.	Explain threats to WLANs.				
	4.7 Explain WLAN security mechanisms.	Explain WLAN security mechanisms.				
	4.8 Explain WLAN support for remote site.	Configure a WLAN to support a remote site.				
	4.9 Explain WLC	Configure a WLC WLAN				

	WLAN management interface and WPA2 PSK authentication. 4.10 Explain how to configure a WLC WLAN to use a VLAN interface, a DHCP server, and WPA2 Enterprise authentication 4.11 Explain how to troubleshoot common wireless configuration issues.	to use the management interface and WPA2 PSK authentication. Explain how to configure a WLC WLAN to use a VLAN interface, a DHCP server, and WPA2 Enterprise authentication Explain how to troubleshoot common wireless configuration issues.				
GENERAL OBJECTIVE: 5.0 Know Routing Protocols						
9	5.1 Explain how routers determine the best path. 5.2 Explain how routers forward packets to the destination. 5.3 Explain basic settings on a router. 5.4 Explain the structure of a routing table. 5.5 Contrast static and	Explain how routers determine the best path. Explain how routers forward packets to the destination. Explain how to configure basic settings on a router. Explain the structure of a routing table. Compare static and	Multimedia Projector, Routers and Switches, Packet Tracer Laptop Internet eNSP	• Configure basic Router configuration	• Guide students to configure basic Router configuration	

	dynamic routing concepts.	dynamic routing concepts.				
GENERAL OBJECTIVE: 6.0 Understand IP Static Routing						
10 - 11	6.1 Explain the command syntax for static routes. 6.2 Explain IPv4 and IPv6 static routes. 6.3 Explain IPv4 and IPv6 default static routes. 6.4 Explain the floating static route to provide a backup connection. 6.5 Explain IPv4 and IPv6 static host routes that direct traffic to a specific host.	Explain the command syntax for static routes. Explain how to configure IPv4 and IPv6 static routes. Explain how to configure IPv4 and IPv6 default static routes. Explain how to configure a floating static route to provide a backup connection. Explain how to configure IPv4 and IPv6 static host routes that direct traffic to a specific host.	Multimedia Projector, Routers and Switches, Packet Tracer Laptop Internet eNSP	• Configure IPv4 and IPv6 Static and Default Routes ○ Configure IPv4 and IPv6 Static and Default Routes ○ Configure and Verify IP and IPv6 Addressing on Routers	Guide students to: • Configure IPv4 and IPv6 Static and Default Routes • Configure IPv4 and IPv6 Static and Default Routes Configure and Verify IP and IPv6 Addressing on Routers	
GENERAL OBJECTIVE: 7.0 Understand Troubleshooting Static and Default Routes						
12	7.1 Explain how a router processes packets when a static route is configured. 7.2 Explain how to find and fix common static and default route configuration	Explain how a router processes packets when a static route is configured. Explain how to troubleshoot common static and default route configuration issues.	Multimedia Projector, Routers and Switches, Packet Tracer eNSP Laptop Internet	Troubleshoot Static and Default Route Troubleshoot IPv4 static routes. Troubleshoot IPv6 static routes.	Guide students to: • Troubleshoot Static and Default Route • Troubleshoot IPv4 static routes. • Troubleshoot IPv6 static	

	issues.			Configure IPv4 static routes. Configure IPv4 default routes. Configure IPv6 static routes		
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NETWORK MONITORING AND TROUBLESHOOTING

PROGRAMME: HIGHER NATIONAL DIPLOMA NETWORKING AND CLOUD COMPUTING		
COURSE: NETWORK MONITORING AND TROUBLESHOOTING	COURSE CODE: NCC 414	Contact Hours: 3 hours/ week
	CREDIT UNITS: 3	Theoretical: 1hour/week
Year: II Semester: I	Pre-requisite:	Practical: 2hours /week
GOAL: The course is designed to enable students monitor and troubleshoot networks		
GENERAL OBJECTIVES: On completion of this course, the students should be able to: 1.0 Understand Troubleshooting and Monitoring concepts 2.0 Know Troubleshooting Tools and Techniques 3.0 Understand Network Traffic Analysis 4.0 Understand Physical Network Troubleshooting 5.0 Understand Virtual and Cloud Network Troubleshooting 6.0 Understand Network Monitoring and Management 7.0 Understand Network Troubleshooting and Monitoring Best Practices and Applications		

PROGRAMME: HIGHER NATIONAL DIPLOMA NETWORKING AND CLOUD COMPUTING						
COURSE: NETWORK TROUBLESHOOTING AND MONITORING		COURSE CODE: NCC 414		Contact Hours: 3hours/week		
		CREDIT UNITS: 3		Theoretical: 1hour/week		
Year: II Semester: I		Pre-requisite:		Practical: 2hours /week		
COURSE SPECIFICATION: THEORETICAL AND PRACTICAL						
GOAL: The course is designed to enable students monitor and troubleshoot networks						
GENERAL OBJECTIVE 1.0: Know Network Troubleshooting and Network Monitoring Concepts						
THEORETICAL CONTENT				PRACTICAL CONTENT		
Week	Specific Learning Outcome	Teacher’s Activities	Resources	Specific Learning Outcome	Teacher’s Activities	Evaluation
1	1.1 Explain Network Troubleshooting and Monitoring concepts 1.2 Explain the importance of effective Network Troubleshooting and Monitoring 1.3 Explain network architecture and components 1.4 Explain Network Troubleshooting process 1.5 Explain common network issues 1.6 Explain various network	• Explain Network Troubleshooting and Monitoring concepts • Explain the importance of Network Troubleshooting and Monitoring • Explain network architecture and components • Explain Network Troubleshooting process • Explain common network problems • Explain various	Marker Board and Marker, Projector			

	protocols (TCP/IP, DNS, DHCP etc.)	network protocols (TCP/IP, DNS, DHCP etc.)				
GENERAL OBJECTIVE 2.0: Understand Troubleshooting Tools and Techniques						
2-3	2.1 Explain types of Network Troubleshooting tools (Command-line, Applications and Hardware tools) 2.2 Explain the usage of the outlined Network Troubleshooting tools and utilities in Windows and Linux Operating System platforms 2.3 Explain ways of interpreting results obtained from troubleshooting tools and utilities	• Explain the Command-line, Applications and Hardware tools involved in Network Troubleshooting • Explain how to use the outlined tools and utilities in Windows and Linux Operating System platforms • Explain results obtained from the Network Troubleshooting tools and utilities	Marker Board and Marker, Projector PC, OS (Windows and Linux), Internet connectivity, Networking toolbox, Ethernet Cable Command-line tools (ping, tracert, nslookup, Nmap, etc.), Packet sniffers (e.g., Wireshark), Port Scanners, Protocol Analyzer, Wi-Fi Analyzer, Bandwidth Speed Tester.	• Install and configure the troubleshooting tools on Windows and Linux Operating System platforms • Carry out Network Troubleshooting using the outlined tools and utilities • Interpret the results obtained from the outlined tools and utilities • Troubleshoot network cables using cable testers • Interpret results obtained from diagnostic tools	• Guide students to install and configure the troubleshooting tools and utilities ○ Guide students to troubleshoot using the outlined tools and utilities ○ Guide students to interpret results obtained from troubleshooting tools and utilities ○ Guide students to troubleshoot	

					t network cables using cable testers	
GENERAL OBJECTIVE 3.0: Understand Network Traffic Analysis						
4	3.1 Explain Network Traffic Analysis 3.2 Explain network traffic capturing and analysis using Wireshark 3.3 Analyse the data captured by Wireshark	• Explain Network Traffic Analysis • Explain network traffic capturing and analysis using Wireshark • Explain how to analyse the data captured by Wireshark	Marker Board and Marker, Projector PC, LAN, Wireshark	• Carry out packet sniffing using Wireshark • Analyse the packets captured and identify the network protocols	• Guide students to capture network traffic using Wireshark • Guide students to analyse captured traffic and identify the network protocols	
GENERAL OBJECTIVE 4.0: Understand Physical Network Troubleshooting						
5- 8	4.1 Explain various network hardware problems 4.2 Explain network device performance and optimization 4.3 Explain network devices troubleshooting process (PCs, Routers, Switches, Firewalls etc.)	• Explain the various network hardware problems ○ Explain network device performance and optimization ○ Explain the process involved in troubleshooting network devices (PCs, Routers, Switches, Firewalls etc.)	Marker Board and Marker, Projector Network devices (Routers, Switches, Ethernet cables, etc.), Network toolkits, LAN, Internet connectivity, Printer, VoIP devices (IP-Phone, softphone, etc.), VoIP Server	• Create a Local Area Network (LAN) using physical devices ○ Create a Local Area Network (LAN) using simulation software (Packet tracer, eNSP, GNS3, etc.) ○ Troubleshoot physical issues such as cable faults, connector	• Guide students to create LAN using physical devices ○ Guide students to create LAN using network simulation software	

	4.4 Identify common network service issues (Internet connectivity, file sharing, printing etc.) 4.5 Explain IP addressing and subnetting 4.6 Explain network device configuration issues and diagnosis 4.7 Explain wireless network technologies 4.8 Explain wireless network optimization and configuration settings 4.9 Explain network security concepts 4.10 Explain Voice and Video over IP (VoIP) technologies and protocols	○ Explain network service issues (Internet connectivity, file sharing, printing etc.) ○ Explain the concepts of IP addressing and subnetting ○ Explain network device configuration issues and how to mitigate them ○ Explain wireless network technologies and their functionalities ○ Explain how to configure and optimize wireless network settings ○ Explain network security concepts ○ Explain Voice and Video over IP (VoIP) technologies and protocols	(Asterisk etc.) Wireless Access point	problems, and electrical interference ○ Identify and resolve frame errors, collisions, and MAC address conflicts ○ Analyse and troubleshoot network switch configurations ○ Configure and troubleshoot router using the various routing protocols (OSPF, RIP, EGP, IS-IS etc.)	○ Guide students to troubleshoot physical problems such as cable faults, connector problems and electrical interference ○ Guide students to identify and resolve frame errors, collisions, and MAC address conflicts ○ Guide students to analyse and troubleshoot switch configurations ○ Guide	
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	4.11 Identify common VoIP issues 4.12 Explain how to Configure and optimize VoIP devices	○ Explain common issues in VoIP and how to mitigate them ○ Explain how to install, configure, and mitigate VoIP devices		○ Troubleshoot IP addressing, routing and subnetting problems ○ Troubleshoot connectivity issues using ping, traceroute/tracert, and IP scanners ○ Analyse TCP and UDP for performance and congestion issues ○ Troubleshoot TCP connection problems such as Packet loss and latency ○ Resolve issues related	students to configure and troubleshoot router using the various routing protocols (OSPF, RIP, EGP, IS-IS etc.) ○ Guide students to troubleshoot IP addressing, routing and subnetting problems ○ Guide students to troubleshoot network connectivity issues using ping, traceroute/tracert, and IP scanners ○ Guide	
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				to DNS, DHCP and HTTP ○ Troubleshoot email and file transfer issues ○ Analyse device logs and error messages ○ Troubleshoot wireless signal interference, coverage, and authentication problems ○ Identify and mitigate common VoIP issues ○ Configure and optimize VoIP	students to analyse TCP and UDP for performance and congestion issues ○ Guide students to troubleshoot TCP connection problems like Packet loss and latency ○ Guide students to identify and resolve issues related to DNS, DHCP and HTTP ○ Guide students to troubleshoot email and file transfer	
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				devices	issues ○ Guide students to access and analyse network device logs and error messages ○ Guide students to troubleshoot wireless signal interference, coverage, and authentication problems ○ Guide students to identify and mitigate VoIP issues ○ Guide students to configure and optimize	
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					VoIP devices	
GENERAL OBJECTIVE 5.0: Understand Virtual and Cloud Network Troubleshooting						
9-11	5.1 Explain Network Virtualization (VMware, VirtualBox, Hyper-V, etc.) 5.2 Explain common issues in virtualized network environment and their causes 5.3 Explain problems related to virtual switches, routers, network overlays and their causes 5.4 Explain virtual network logs and performance metrics 5.5 Explain networking in cloud environments 5.6 Explain causes of network connectivity and performance issues in cloud environments	• Explain Network Virtualization (VMware, VirtualBox, Hyper-V etc.) • Explain common issues in virtualized network environment and their causes • Explain problems related to virtual switches, routers, network overlays and their causes • Explain how to access and analyse logs and performance metrics in virtual environments • Explain networking cloud environments • Explain causes of network connectivity and performance issues in cloud environments	Marker Board and Marker, PCs, Internet connectivity, Hypervisor (VMware, VirtualBox, Hyper-V etc.) Cloud platform (AWS, Azure etc.)	• Set up virtual environments and install virtual machines • Set up LAN within the virtual environment • Troubleshoot virtual network configurations and connectivity issues • Access and analyse log files and performance metrics in virtual environments • Set up LAN in cloud environments	• Guide students to install Hypervisor and set up virtual machines • Guide students to set up Local Area Network (LAN) in cloud environment • Guide students to troubleshoot virtual network configurations and connectivity issues • Teach the students how to	

	5.7 Explain cloud network logs and metrics 5.8 Explain issues related to cloud network configuration	• Explain how to access and analyse logs and performance metrics • Explain issues related to cloud network configuration		• Troubleshoot network connectivity in cloud environment • Access and analyse cloud network logs and performance metrics • Troubleshoot configuration issues within cloud environment	access and analyse log files on virtual machines • Guide students to set up Local Area Network in a cloud environment • Guide students to troubleshoot network connectivity issues in cloud environment • Teach students how to access and analyse network logs and performance metrics in	
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					cloud environments	
					Guide students to troubleshoot configuration issues in cloud environments	
GENERAL OBJECTIVE 6.0: Understand Network Monitoring and Management						
12-13	6.1 Explain Network Monitoring and Management 6.2 Explain Network Monitoring and Management tools (SNMP, NMS etc.) 6.3 Explain Network performance metrics and monitoring techniques 6.4 Explain network performance optimization	- Explain Network Monitoring and Management - Explain Network Monitoring and Management tools (SNMP, NMS etc.) - Explain network performance metrics and monitoring techniques - Explain network performance optimization	Marker Board and Marker PCs, LAN, Internet connectivity, Network Monitoring applications (SolarWinds, Datadog, Manage Engine etc.)	- Install and configure Network Monitoring systems - Monitor network performance using SNMP, NetFlow and Syslog - Analyse network performance metrics and identify bottlenecks	- Guide students to install and configure Network Monitoring systems - Guide students to monitor network performance using SNMP, NetFlow and Syslog - Guide students to analyse	

				○ Optimize network performance through configuration changes	network performance metrics and identify bottlenecks ○ Guide students to optimize network performance through configuration changes	
GENERAL OBJECTIVE 7.0: Understand Network Troubleshooting and Monitoring Best Practices and Applications						
14-15	7.1 Explain effective network monitoring implementation strategies 7.2 Explain the procedures to configure alerts and notifications for critical network events 7.3 Explain troubleshooting methodologies and techniques and their applications in solving complex problems	• Explain effective network monitoring implementation strategies • Explain the procedures to configure alerts and notifications for critical network events • Explain troubleshooting methodologies and techniques and their applications in solving complex problems	Marker Board and Marker PCs, LAN, Internet connectivity, Network diagram tools (Microsoft Visio, Lan Flow, SolarWinds, Intermapper, Creately, Smart Draw, etc.)	• Configure alerts and notifications for critical network events • Analyse network monitoring data to proactively identify issues • Apply troubleshooting methodologies and techniques to solve	• Guide students to configure alerts and notifications for critical network events • Guide students to analyse network monitoring data to proactively identify issues	

	7.4 Explain the importance of network documentation and reporting 7.5 Explain network diagram	• Explain network documentation and reporting, importance, and real-world application. • Explain network diagrams		complex problems • Create network diagrams • Write effective network troubleshooting and monitoring reports • Analyse real-world network troubleshooting scenarios	• Guide students to apply troubleshooting methodologies and techniques to solve complex problems • Guide students to design network diagrams using Network Diagram tools (Microsoft Visio, SolarWinds, Intermapper, Lan Flow, Smart Draw etc.) • Guide students to write effective	
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				• Implement effective network monitoring strategies	network troubleshooting and monitoring reports • Guide students using case studies to implement effective network monitoring strategies	
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YEAR TWO SEMESTER TWO COURSES

CLOUD COMPUTING II

PROGRAMME: HIGHER NATIONAL DIPLOMA (HND) NETWORKING AND CLOUD COMPUTING			
COURSE: CLOUD COMPUTING II	CODE: NCC 421	Credit Unit: 3	CONTACT HOURS: 3 HOURS/WEEK
YEAR: 11 SEMESTER: 11	PRE-REQUISITE:		THEORETICAL: 1 HOURS/WEEK PRACTICAL: 2 HOURS/WEEK
GOAL: This course is designed to equip students with Cloud Services Implementation			
GENERAL OBJECTIVES On completion of the course, the student should be able to: 1.0 Understand Cloud Architecture 2.0 Understand the implementation of Compute in the Cloud. 3.0 Understand the implementation of storage in the Cloud. 4.0 Understand the implementation of Database in the cloud. 5.0 Understand Automatic Scaling and Monitoring.			

PROGRAMME: HIGHER NATIONAL DIPLOMA IN NETWORKING AND CLOUD COMPUTING						
COURSE: CLOUD COMPUTING II			CODE: NCC 421	CONTACT HOURS: 3 HOURS/WEEK		
Credit Unit: 3				Theoretical: 1 HOUR/WEEK Practical: 2 HOURS/WEEK		
Year: II Semester: II						
GOAL: This course is designed to equip students to attain foundational comprehension of cloud implementation.						
Course Specification: THEORETICAL CONTENT				PRACTICAL CONTENT		
GENERAL OBJECTIVE 1.0: Understand cloud architecture.						
Week	Specific Learning Outcome	Teachers’ Activities	Learning Resources	Specific Learning Outcome	Teachers’ Activities	Evaluation
1-3	1.1 Explain the AWS Well-Architected Framework, including the five pillars. 1.2 Explain the design principles of the AWS Well-Architected Framework. 1.3 Explain the importance of reliability and high availability. 1.4 Explain how AWS Trusted Advisor helps customers. 1.5 Explain AWS Trusted Advisor recommendations.	• Explain the AWS Well-Architected Framework, including the five pillars. • Explain the design principles of the AWS Well-Architected Framework. • Explain the importance of reliability and high availability. • Explain how AWS Trusted Advisor helps customers. • Explain AWS Trusted Advisor recommendations	AWS platforms, Multi-media Projector, Lecture notes, Internet Computers, etc.			Explain the AWS Well-Architected Framework, including the five pillars.

General Objective 2.0: Understand the implementation of compute in the cloud.						
4-6	2.1 Explain an overview of different compute services in the cloud 2.2 Explain the reasons for using cloud compute 2.3 Explain the functionality in the compute console 2.4 Explain basic functions in cloud compute to build a virtual computing environment 2.5 Explain compute cost optimization elements 2.6 Explain when to use platform as a service (PaaS) 2.7 Explain when to use serverless compute	• Explain an overview of different compute services in the cloud • Explain the reasons for using cloud compute • Explain the functionality in the compute console • Explain basic functions in compute to build a virtual computing environment • Explain compute cost optimization elements • Explain when to use PaaS using such examples such AWS Elastic Beanstalk, Azure App Service, or any other PaaS • Explain when to use	Internet, AWS platform or Azure Platform, or any other platform Multimedia Projector, Computer, Lecture notes, Tutorials, Marker, Markerboard, Books/Journals, PowerPoint slides	• Sign up for an account • Explore the features of AWS EC2, Azure App, or any other platform • Create virtual machines • Configure access and security • Attach storage and take snapshots • create custom images •	Guide students to: • Sign up for an account • Explore the features of AWS EC2, Azure App, or any other platform • Sign up for an account if you don't have one, • Create virtual machines • Configure access and security • Configure load scaling and load	Explain an overview of different compute services in the cloud Explain the reasons for using cloud compute

	service	serverless compute services such as AWS Lambda, Azure Functions, or any other			balancing	
	2.8 Explain how to run containerized applications in a cluster of managed servers	Explain how to run containerized applications in a cluster of managed servers			- Attach storage and take snapshots - Create custom images	
GENERAL OBJECTIVE 3.0: Understand the implementation of storage in the cloud.						
7-9	3.1 Explain different storage services in the cloud 3.2 Explain the functionality in cloud storage services 3.3 Explain functions in cloud storage service to build a storage solution 3.4 Explain cloud file system 3.5 Explain the functionality in cloud file system	- Explain the different types of storage - Explain the functionality in cloud storage services using Amazon EBS, Azure Blob Storage, and any other storage service - Explain functions in cloud storage service to build a storage solution such Amazon EBS to build an Amazon EC2 storage solution, or Azure Blob Storage - Explain cloud file system such as Amazon EFS, Azure Files, or any other	Internet, AWS platform or Azure Platform, or any other platform Multimedia Projector, Computer, Lecture notes, Tutorials, Marker, Markerboard, Books/Journals, PowerPoint	Sign into the created account to explore the features of AWS EBS, Azure Blob Storage, or any other storage service to create, attach, and format volumes, take snapshots and create backup, resize volumes, enable encryption for the volumes, configure high availability and redundancy, create custom images and templates	Guide students on how to sign into the created account in order to explore the features of AWS EBS, Azure Blob Storage, or any other storage service to create, attach, and format volumes, take snapshots and create backup, resize volumes,	Explain the functionality in cloud storage services

	3.6 Explain cloud archiving system 3.7 Explain the functionality in cloud archiving system 3.8 Explain the difference between different cloud storage and archiving services	• Explain the functionality in cloud file system such as Amazon EFS, Azure Files, or any other • Explain cloud archiving system such as Amazon S3 Glacier, Azure Archive Storage, or any other • Explain the functionality in cloud archiving system • Explain the difference between different cloud storage and archiving services Amazon EBS, Amazon S3, Amazon EFS, and Amazon S3 Glacier; Azure Blob Storage, Azure Managed Disks, Azure Files, Azure Archive Storage	slides		enable encryption for the volumes, configure high availability and redundancy, create custom images and templates	
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GENERAL OBJECTIVE 4.0: Understand the implementation of database in the cloud.

10-12	4.1 Explain relational database use in cloud services 4.2 Explain the functionality in cloud base database 4.3 Explain NoSQL database use in cloud services 4.4 Explain the functionality in NoSQL database 4.5 Explain data warehousing services	• Explain relational database use in cloud services such Amazon Relational Database Service (Amazon RDS), Azure SQL database, and any other cloud database service • Explain the functionality in cloud based data using Amazon RDS, Azure SQL Database • Explain NoSQL database use in the cloud such as Amazon DynamoDB, Azure Cosmos DB • Explain the functionality in NoSQL database such as Amazon DynamoDB, Azure Cosmos DB, or any other • Explain cloud-based data warehousing services such as	Internet, AWS platform or Azure Platform, or any other platform Multimedia Projector, Computer, Lecture notes, Tutorials, Marker, Markerboard, Books/Journal s, PowerPoint slides	• Build a DB Server and Interact with the DB Using an App	• Guide the students to build a DB Server and Interact with the DB Using an App	Explain relational database use in cloud services
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	4.6 Explain relational database engine 4.7 Explain the tasks required in launching, configuring, and interacting with a cloud database	Amazon Redshift, Azure Synapse Analytics, or any other • Explain relational database engine such as Amazon Aurora, Azure Database for Hyperscale • Explain tasks in an RDS database, such as launching, configuring, and interacting				
GENERAL OBJECTIVE 5.0: Understand automatic scaling and monitoring						
13-15	5.1 Explain how to distribute traffic across Amazon Elastic Compute Cloud (Amazon EC2) instances by using Elastic Load Balancing 5.2 Explain how Amazon CloudWatch enables you to monitor AWS resources and applications in real time 5.3 Explain how Amazon EC2 Auto	• Explain how to distribute traffic across Amazon Elastic Compute Cloud (Amazon EC2) instances by using Elastic Load Balancing • Explain how Amazon CloudWatch enables you to monitor AWS resources and applications in real time • Explain how Amazon EC2 Auto Scaling launches and releases	Internet, AWS Platform, Markers, Marker board, Projector, Lecture notes, etc.	• Perform scaling and load balancing	• Guide the students to perform scaling and load balancing	Explain how to distribute traffic across Amazon Elastic Compute Cloud (Amazon EC2) instances by using Elastic Load Balancing

	Scaling launches and releases servers in response to workload changes	servers in response to workload changes				
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ASSESSMENT: The continuous assessment; tests, quizzes will be awarded 40% of the total score. The end of the Semester Examination will make up for the remaining 60% of the total score.

ENTERPRISE NETWORKING, SECURITY, AND AUTOMATION

PROGRAMME: HIGHER NATIONAL DIPLOMA NETWORKING AND CLOUD COMPUTING		
COURSE: ENTERPRISE NETWORKING, SECURITY, AND AUTOMATION	COURSE CODE: NCC 422	Contact Hours: 4 hours/week
	Credit units: 4	Theoretical: 1 hours /week
Year: II Semester: II	Pre-requisite:	Practical: 3 hours /week
GOAL: This Course is designed to equip students with skills in Enterprise Networking, Security, and Automation.		
GENERAL OBJECTIVES: on completion of this of the course, the students should be able to; 1.0 Understand Single-Area Open Shortest Path First (OSPFv2) 2.0 Understand Network Security Concepts 3.0 Understand Access Control List (ACL) Concepts and Configuration 4.0 Understand Network Address Translation (NAT) for IPv4 Wide Area Network (WAN) Concepts 5.0 Understand Wide Area Network (WAN) Concepts 6.0 Understand Virtual Private Network (VPN) and IPsec Concepts 7.0 Know Concept of Quality of Service (QoS) 8.0 Understand Network Management 9.0 Understand Network Design 10. Understand Network Troubleshooting 11. Understand Network Virtualization 12. Understand Network Automation.		

PROGRAMME: HIGHER NATIONAL DIPLOMA NETWORKING AND CLOUD COMPUTING							
COURSE: ENTERPRISE NETWORKING, SECURITY, AND AUTOMATION				COURSE CODE: NCC 422		Contact Hours: 4 hours/week	
				Credit unit: 4		Theoretical: 1 hours /week	
Year: II Semester: II				Pre-requisite:		Practical: 3 hours /week	
GOAL: This Course is designed to equip with skills in enterprise networking, security, and automation							
GENERAL OBJECTIVE 1.0: Understand Single-Area Open Shortest Path First (OSPFv2)							
THEORETICAL CONTENT				PRACTICAL CONTENT			
Week	Specific Learning Outcome	Teacher’s Activities	Resources	Specific Learning Outcome	Teacher’s Activities	Evaluation	
1	1.1 Explain OSPF operations 1.2 Explain the OSPF packet types used in single-area OSPF 1.3 Explain single-area OSPF operations	• Explain OSPF operations • Explain the OSPF packet types used in single-area OSPF • Explain single-area OSPF operations	Multimedia Projector, Routers and Switches, Packet Tracer or eNSP or GNS3, Laptop Internet Books, Journals, Markerboard, Marker, Textbooks	• Configure: • Single-area OSPFv2 in both point-to-point and • Broadcast multi-access networks. • OSPF interface priority to influence a multi-access network's DR/BDR election. • OSPF to propagate a default route • Modify the configuration to change	• Guide students to configure: • Single-area OSPFv2 in both point-to-point and • Broadcast multi-access networks. • OSPF interface priority to influence a multi-access network's DR/BDR election. • Verify a single-area OSPFv2 implementation	Explain single-area OSPF operations	
					OSPF to propagate a default route		

				the operation of single area OSPFv2	Modify the configuration to change the operation of single area OSPFv2	
GENERAL OBJECTIVE: 2.0 Understand Network Security Concepts						
2-3	2.1 Explain network vulnerabilities, threats, and mitigation techniques 2.2 Explain the current state of cybersecurity and vectors of data loss 2.3 Explain the tools used by threat actors to exploit networks 2.4 Explain malware types 2.5 Explain common network attacks	- Explain network vulnerabilities, threats, and mitigation techniques - Explain the current state of cybersecurity and vectors of data loss - Explain the tools used by threat actors to exploit networks - Explain malware types - Explain common network attacks	Multimedia Projector, Computer, Internet, Books, Journals, Markerboard, Marker, Textbooks, wireshark	Explore DNS Traffic using Wireshark	Guide the students to explore DNS Traffic using Wireshark	Explain the current state of cybersecurity and vectors of data loss

	2.6 Explain IP vulnerabilities and exploitations 2.7 Explain TCP and UDP vulnerabilities and mitigations 2.8 Explain common cryptographic processes used to protect data in transit 2.9 Explain Social Engineering techniques	• Explain IP vulnerabilities and exploitations • Explain TCP and UDP vulnerabilities and mitigations • Explain common cryptographic processes used to protect data in transit • Explain Social Engineering techniques				
• GENERAL OBJECTIVE: 3.0 Understand ACL Concepts and Configuration						
4-5	3.1 Explain how ACLs are used as part of a network security policy. 3.2 Explain how ACLs filter traffic.	• Explain how ACLs are used as part of a network security policy. • Explain how ACLs filter traffic.	Multimedia Projector, Routers and Switches, Packet Tracer or eNSP or GNS3, Laptop Internet Multimedia Projector, Routers and	Configure: Numbered Standard IPv4 ACLs Extended IPv4 ACLs - Configure Extended • Modify Standard IPv4 ACLs- Configurations	Guide the students to configure: Numbered Standard IPv4 ACLs Extended IPv4 ACLs - Configure Extended • Modify Standard IPv4	Explain how ACLs filter traffic.

	3.3 Explain how ACLs use wildcard masks. 3.4 Explain how to create ACLs. 3.5 Contrast standard and extended IPv4 ACLs	- Explain how ACLs use wildcard masks. - Explain how to create ACLs. Compare standard and extended IPv4 ACLs Explain the implementation of IPv4 ACLs to filter traffic and secure administrative access. Explain how to configure standard IPv4 ACLs to filter traffic to meet networking requirements. Explain the use of sequence numbers to edit existing standard IPv4 ACLs. 3.10 Configure a standard ACL to secure Vty access. 3.11 Configure extended IPv4 ACLs to	Switches, Packet Tracer or eNSP or GNS3, Laptop Internet	Apply IPv4 ACLs to filter traffic and secure administrative access. Set up standard IPv4 ACLs to filter traffic to meet networking requirements. Apply sequence numbers to edit existing standard IPv4 ACLs. Set up a standard ACL to secure vty access. Set up extended IPv4 ACLs to filter traffic according to networking requirements.	ACLs- Configurations	
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		filter traffic according to networking requirements.				
GENERAL OBJECTIVE: 4.0 Understand NAT and PAT for IPv4 and IPv6						
6	4.1 Explain the purpose and function of NAT 4.2 Explain the operation of different types of NAT. 4.3 Explain the advantages and disadvantages of NAT 4.5 Explain NAT for IPv6.	- Explain NAT services on edge router to provide IPv4 address scalability. - Explain the purpose and function of NAT - Explain the operation of different types of NAT. - Explain the advantages and disadvantages of NAT - Explain NAT for IPv6.	Multimedia Projector, Routers and Switches, Packet Tracer or eNSP or GNS3, Laptop Internet Books, Journals, Markerboard, Marker, Textbooks	Configure: Static NAT Dynamic NAT and PAT NAT for IPv4 Set up static NAT using the CLI.	Guide the students to configure: Static NAT Dynamic NAT and PAT NAT for IPv4 Set up static NAT using the CLI.	Explain the advantages and disadvantages of NAT
GENERAL OBJECTIVE: 5.0 Understand WAN Concepts						
7	5.1 Explain how WAN access technologies can be used to satisfy business requirements	Explain how WAN access technologies can be used to	Multimedia Projector, Laptop Internet Books, Journals, Markerboard,	Conduct a Research on Broadband Internet Access Technologies WAN Concepts	Guide the students to Research Broadband Internet Access Technologies WAN Concepts	Explain how WANs operate.

	5.2 Explain the purpose of WAN 5.3 Explain how WANs operate. 5.4 Differentiate between traditional WAN connectivity, modern WAN connectivity, and internet-based WAN connectivity options.	satisfy business requirements • Explain the purpose of a WAN • Explain how WANs operate. • Discuss the differences between traditional WAN connectivity, modern WAN connectivity, and internet-based WAN connectivity options.	Marker, Textbooks, TeraTerm, Putty			
• GENERAL OBJECTIVE: 6.0 Understand VPN and IPsec Concepts						
8	6.1 Explain how VPNs and IPsec secure site-to-site and remote access connectivity. 6.2 Explain the benefits of VPN technology. 6.3 Explain different types of VPNs	• Explain how VPNs and IPsec secure site-to-site and remote access connectivity. • Explain the benefits of VPN technology.	Multimedia Projector, Laptop Internet Books, Journals, Markerboard, Marker, Textbooks.t	Configure site to site VPN Configure remote VPN	Guide students to: Configure site to site VPN Configure remote VPN	Explain different types of VPNs

	6.4 Explain how the IPsec framework is used to secure network traffic.	- Explain different types of VPNs - Explain how IPsec framework is used to secure network traffic.				
GENERAL OBJECTIVE: Know Concepts of quality of service						
9	7.1 Explain how networking devices implement QoS 7.2 Explain how network transmission characteristics impact QoS	- Explain how networking devices implement QoS - Explain how network transmission characteristics impact QoS				Explain how networking devices implement QoS
GENERAL OBJECTIVE: 8.0 Understand Network Management						
10	8.1 Explain protocols used to manage a network. Explain the use of LLDP to map network topology.	- Explain protocols used to manage a network. - Explain the essence of mapping network topology.	Multimedia Projector, Routers and Switches, Packet Tracer or eNSP or GNS3, Laptop Internet, TeraTerm, Putty	Configure: CDP, LLDP, and NTP Use CDP to Map a Network Use LLDP to Map a Network Use IOS and VRP commands to back up and restore an configuration	Show the students how to Configure: CDP, LLDP, and NTP Use CDP to Map a Network Use LLDP to Map a Network Guide the students to:	Explain syslog operation.

	8.2 Explain SNMP operation. Explain the implementation of NTP between an NTP client and an NTP server. 8.3 Explain syslog operation. 8.4 Explain Upgrade of an IOS and VRP system image.	• Explain SNMP operation. • Explain the implementation of NTP between an NTP client and an NTP server. • Explain syslog operation. • Explain IOS and VRP operations.		file. • Verify NTP Research Network Monitoring Software Back-Up Use TeraTerm or Putty to Manage Router Configuration Files Use TFTP, Flash and USB to Upgrade an IOS and VRP system image.	verify NTP Research Network Monitoring Software Back-Up Use TeraTerm or Putty to Manage Router Configuration Files Use TFTP, Flash, and USB to Manage Upgrade an IOS and VRP system image.	
GENERAL OBJECTIVE 9.0: Understand Network Design						
11	9.1 Explain the characteristics of scalable network architectures. 9.2 Explain how data, voice, and video are converged in a switched network. 9.3 Explain considerations for designing a scalable network.	• Explain the characteristics of scalable network architectures. • Explain how data, voice, and video are converged in a switched network. • Explain considerations for designing a scalable network.	Multimedia Projector, Cisco Routers and Switches, Packet Tracer Laptop Internet			Explain considerations for designing a scalable network.

	9.4 Explain how switch hardware features support network requirements. 9.5 Explain the types of routers available for small-to-medium-sized business networks.	• Explain how switch hardware features support network requirements. • Explain the types of routers available for small-to-medium-sized business networks.				
GENERAL OBJECTIVE: 10 Understand Network Troubleshooting						
12	10.1 List the symptoms and causes of network problems using a layered model. 10.2 Explain how to find and fix a network using the layered model.	• Determine the symptoms and causes of network problems using a layered model. • Explain how to troubleshoot a network using the layered model.	Multimedia Projector, Cisco Routers and Switches, Packet Tracer Laptop Internet	Troubleshoot Enterprise Networks Document the Network Use Documentation to Solve Issues Find and fix a network using the layered model.	Guide students to: Troubleshoot Enterprise Networks Troubleshooting Challenge - Document the Network Troubleshooting Challenge - Use Documentation to Solve Issues	Explain how to fix a network using the layered model.
GENERAL OBJECTIVE:11.0 Understand Network Virtualization						
13	11.1 Explain the importance of cloud computing. 11.2 Explain the virtualization of network devices and services.	• Explain the importance of cloud computing. • Explain the virtualization of network	Multimedia Projector, Routers and Switches, Packet Tracer or eNSP Laptop Internet Virtualbox	Install Linux in a Virtual Machine and Explore the GUI	Guide students to install Linux in a Virtual Machine and Explore the GUI	Explain the importance of cloud computing.

	11.3 Explain software-defined networking. 11.4 Explain controllers used in network programming	devices and services. • Explain software-defined networking. • Explain controllers used in network programming				
GENERAL OBJECTIVE: 12.0 Understand Network Automation						
14-15	12.1 Explain how network automation is enabled through RESTful APIs and Configuration management tools. 12.2 Explain automation. 12.3 Contrast JSON, YAML, and XML data formats.	• Explain how network automation is enabled through RESTful APIs and Configuration management tools. • Explain automation. • Compare JSON, YAML, and XML data formats.	Multimedia Projector, Routers and Switches, Packet Tracer or eNSP Laptop Internet Virtualbox	•	•	• Explain how REST enables computer-to-computer communications.

	12.4 Explain how APIs enable computer-to-computer communications. 12.5 Explain how REST enables computer-to-computer communications. 12.6 Contrast the configuration management tools Puppet, Chef, Ansible, and SaltStack 12.7 Explain how Cisco DNA centre enables intent-based networking	• Explain how APIs enable computer-to-computer communications. • Explain how REST enables computer-to-computer communications. • Compare the configuration management tools Puppet, Chef, Ansible, and SaltStack • Explain how Cisco DNA center enables intent-based networking				
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ETHICAL AND PROFESSIONAL PRACTICE IN NETWORKING AND CLOUD COMPUTING

PROGRAMME: HIGHER NATIONAL DIPLOMA NETWORKING AND CLOUD COMPUTING		
COURSE TITLE: Ethical and Professional Practice In networking and cloud computing	Course Code: NCC 423	CONTACT HOURS: 2 HOURS/week
YEAR: 11 SEMESTER 11	PRE-REQUISITE:	Theoretical: 2 HOURS/week
	CREDIT UNIT: 2	Practical: 0 HOURS/week
GOAL: This course is designed to enable students to acquire knowledge and skills to navigate the ethical challenges that arise in the field of Networking and Cloud Computing		
General Objectives: On completion of the course, the students should be able to: 1.0 Understand ethical Awareness related to the design, development, and deployment of Networking and Cloud Computing Technologies. 2.0 Understand ethical decision-making when working with Networking and Cloud Computing Technologies. 3.0 Understand how to design Networking and Cloud Computing systems that adhere to ethical principles. 4.0 Comprehend the importance of safeguarding user privacy and personal data in Networking and Cloud Computing applications. 5.0 Explore the societal and legal implications of Networking and Cloud Computing Technologies. 6.0 Cultivate professional ethics and responsible conduct in Networking and Cloud Computing development and research 7.0 Explore the ethics of Networking and Cloud Computing for Intelligence Analysis		

PROGRAMME: HIGHER NATIONAL DIPLOMA NETWORKING AND CLOUD COMPUTING						
COURSE: Ethical Professional Practice in Networking and cloud computing			COURSE CODE: NCC 423		CONTACT HOURS: 2 hrs.	
GOAL: This course is designed to enable students to acquire knowledge and skills to navigate the ethical challenges that arise in the field of Networking and Cloud Computing						
Year: II Semester: II						
COURSE SPECIFICATION: Theoretical Contents:				Practical Contents:.		
General Objective 1.0: Understand ethical Awareness related to the design, development, and deployment of Networking and Cloud Computing technologies						
Week	Specific Learning Outcomes	Teachers Activities	Learning Resources	Specific Learning Outcomes	Teachers Activities	Evaluation
1	1.1 Explain Ethical Principles in Networking and Cloud Computing 1.2 Explain Ethical Dilemmas in Networking and Cloud Computing Applications 1.3. Explain the Impact of on Society and Individuals 1.4. Explain Privacy and Data Protection in Networking and Cloud Computing	• Explain principles and values relevant to Networking and Cloud Computing development. • Discuss ethical dilemmas in Networking and Cloud Computing applications. • Explain Networking and Cloud Computing’s societal impact. • Discuss privacy and data protection regulations in Networking and	Text books, multi-media Projector, Computer, marker, markerboard, Lecture notes			Explain Ethical Dilemmas in Networking and Cloud Computing Applications

	1.5 Explain how to evaluate Bias and Fairness in Networking and Cloud Computing	Cloud Computing. • Discuss Bias and Fairness in Networking and Cloud Computing				
GENERAL OBJECTIVE 2.0: Understand ethical decision-making when working with Networking and Cloud Computing Technologies.						
4	2.1 Explain ethical Decision-making in Networking and Cloud Computing 2.2. Analyze Ethical Dilemmas in Networking and Cloud Computing Applications 2.3. Explain ethical Considerations in Networking and Cloud Computing Research 2.4. Explain Impact of Stakeholders in Networking and Cloud Computing Decisions 2.5. Explain how to apply Ethical Frameworks in Networking and Cloud Computing Decision-making	• Explain ethical decision-making principles and methodologies. • Discuss ethical dilemmas in Networking and Cloud Computing applications. • Discuss ethical considerations in Networking and Cloud Computing research projects. • Discuss the impact of Networking and Cloud Computing decisions on stakeholders. • Discuss ethical frameworks applicable to Networking and Cloud Computing	Text books, multi-media Projector, Computer, marker, markerboard, Lecture notes			Explain ethical decision-making principles and methodologies.

		decisions.				
General Objective 3.0: Understand how to Design Networking and Cloud Computing systems that adhere to ethical principles.						
6	3.1 Explain Ethical Principles in Networking and Cloud Computing Development 3.2. Explain how to Promote Fairness and Transparency in Networking and Cloud Computing 3.3. Describe how to ensure Accountability in Networking and Cloud Computing Decision-making 3.4. Explain ethical Data Collection and Usage in Networking and Cloud Computing	• Explain Ethical Principles in Networking and Cloud Computing Development • Discuss how to promote Fairness and Transparency in Networking and Cloud Computing • Explain Accountability in Networking and Cloud Computing Decision-making • Explain ethical Data Collection and Usage in	Text books, multi-media Projector, Computer, marker, markerboard, Lecture notes			Discuss how to promote Fairness and Transparency in Networking and Cloud Computing
General Objectives: 4.0 Comprehend the importance of safeguarding user privacy and personal data in Networking and Cloud Computing applications.						
10	4.1. Explain Privacy and Data Protection Principles 4.2. Explain how to comply with Privacy Regulations in Networking and Cloud Computing 4.3. Explain how to ensure	• Explain Privacy and Data Protection Principles • Explain Privacy Regulations in Networking and Cloud Computing • Explain Data	Text books, multi-media Projector, Computer, marker, markerboard, Lecture notes			Explain ethical use of User Data in Networking and Cloud Computing Applications

	Data Minimization and Consent in Networking and Cloud Computing 4.4. Explain how to secure Data Storage and Transmission in Networking and Cloud Computing 4.5. Explain ethical use of User Data in Networking and Cloud Computing Applications	Minimization and Consent in Networking and Cloud Computing • Explain how to Secure Data Storage and Transmission in Networking and Cloud Computing • Explain the ethical Use of User Data in Networking and Cloud Computing Applications				
General Objective 5.0: Explore the societal and legal implications of Networking and Cloud Computing technologies						
12	5.1. Explain the Social Impact of Networking and Cloud Computing 5.2 Analyze Legal Implications of Networking and Cloud Computing 5.3 Examine Networking and Cloud Computing Impact on Privacy Laws 5.4. Explain ethical Considerations in AI Applications	• Explain the Social Impact of Networking and Cloud Computing • Analyze Legal Implications of Networking and Cloud Computing • Examine Networking and Cloud Computing Impact on Privacy Laws • Explain ethical Considerations in	Text books, multi-media Projector, Computer, marker, markerboard, Lecture notes			Explain ethical Considerations in Networking and Cloud Computing Applications

	5.5. Explain how to address Networking and Cloud Computing Impact on Human Rights	Networking and Cloud Computing Applications • Discuss how to address Networking and Cloud Computing Impact on Human Rights				
General Objective 6.0 Cultivate professional ethics and responsible conduct in Networking and Cloud Computing development and research						
13	6.1. Explain Professional Ethics in Networking and Cloud Computing 6.2. Explain how to promote Responsible Networking and Cloud Computing Development 6.3. Explain how to Adhere to Ethical Guidelines in Networking and Cloud Computing Research 6.4. Explain transparency and accountability in Networking and Cloud Computing Development 6.5. Explain ethical Decision-making in Networking and Cloud Computing Applications	• Explain Professional Ethics in Networking and Cloud Computing • Explain how to promote responsible Networking and Cloud Computing development • Explain Ethical Guidelines in Networking and Cloud Computing Research • Explain transparency and accountability in Networking and Cloud Computing Development • Explain ethical Decision-making in	Text books, multi-media Projector, Computer, marker, markerboard, Lecture notes			Explain transparency and accountability in Networking and Cloud Computing Development

		Networking and Cloud Computing Applications				
General Objective 7.0: Explore the Ethics of Networking and Cloud Computing for intelligence Analysis						
14	7.1 Explain the ethics of vulnerability testing 7.2 Explain the ethics of visualizations 7.3 Describe the ethics of managing data Analytics 7.4 Describe the ethics of IoT Vulnerabilities	• Describe the ethics of vulnerability testing • Explain the ethics of visualizations • Describe the ethics of managing data Analytics • Describe ethics of IoT Vulnerabilities	Text books, multi-media Projector, Computer, marker, markerboard, Lecture notes			

INTERNET OF THINGS (IOT)

PROGRAMME: HIGHER NATIONAL DIPLOMA NETWORKING AND CLOUD COMPUTING		
COURSE: Internet of Things (IoT)	COURSE CODE: NCC 424	Contact Hours: 3 Hours/week
	Credit units: 3	Theoretical: 1 hour /week
YEAR 11 SEMESTER 11	Pre-requisite:	Practical: 2 hours /week
GOAL: This course is designed to provide student with a Knowledge on Internet of Things (IoT), its areas of application, and significance in modern Technology		
GENERAL OBJECTIVES: On completion of the course, the students should be able to: 1.0 Understand the concept and significance of Internet of Things (IoT). 2.0 Know the importance of security and privacy in IoT systems and some key IoT technologies. 3.0 Understand advanced communication technologies, IoT gateways' challenges in various industrial scenarios, and the capabilities and solutions offered by home gateways. 4.0 Know IoT platforms, their positioning, capabilities, development process, and the role of operating systems 5.0 Understand essential concepts for device development, sensor technologies, the kernel framework and APIs of Lite OS, and AT commands.		

PROGRAMME: HIGHER NATIONAL DIPLOMA NETWORKING AND CLOUD COMPUTING						
COURSE: Internet of Things (IoT)		COURSE CODE: NCC 424			Contact Hours: 3 Hours/week	
		Credit units: 3			Theoretical: 1 hour /week	
Year: I1 Semester: I1		Pre-requisite:			Practical: 2 hours /week	
COURSE SPECIFICATION: THEORETICAL				PRACTICAL		
GOAL: This course is designed to provide students with a knowledge on Internet of Things (IoT), its area of application, and significance in modern Technology						
GENERAL OBJECTIVE 1.0: Understand the concept and significance of Internet of Things (IoT).						
THEORETICAL CONTENT				PRACTICAL CONTENT		
Week	Specific Learning Outcome	Teacher’s Activities	Resources	Specific Learning Outcome	Teacher’s Activities	Evaluation
1-4	1.1 Explain the concept and development history of the IoT, layers of the IoT architecture, and solutions 1.2 Explain IoT industry applications and solutions, covering smart city, smart campus, smart grid, IoV, and industrial IoT scenarios. 1.3 Describe the difficulties facing each industry and the transformation and benefits brought by IoT technology.	- Describe the concept and development history of the IoT, layers of the IoT architecture, and solutions - Describe IoT industry applications and solutions, covering smart city, smart campus, smart grid, IoV, and industrial IoT scenarios - Explain the difficulties facing each industry and the transformation and benefits brought by IoT	Text books, multi-media Projector, Computer, marker, markerboard, Lecture notes, Lab guide, IoT simulation tools, microcontroller’s electronic devices, project development board, Proteus, KiCad.	- Design and model IoT solutions using simulation tools - Design and prototype IoT solutions using electronics and microcontrollers .	- Demonstrate and model IoT solutions using simulation tools - Demonstrate and prototype IoT solutions using electronics and microcontroller s	Explain the concept and development history of the IoT, layers of the IoT architecture, and solutions

	1.4 Describe the combination of the IoT with six emerging technologies and know the development trends of the IoT industry. 1.5 Explain Sensors, Actuators and microcontrollers	technology. • Explain the combination of the IoT with six emerging technologies and know the development trends of the IoT industry. • Describe Sensors, Actuators and microcontrollers				
GENERAL OBJECTIVE: 2.0 Understand the importance of security and privacy in IoT systems and some key IoT systems and Technologies.						
5 – 7	2.1 Describe the security challenges faced by the IoT industry, and list IoT security solutions and related applications. 2.2 Describe common network communications technologies of IoT based on the solutions described in 2.1 2.3 Define key technologies and corresponding solutions of the widely	• Explain the security challenges faced by the IoT industry, and introduces some IoT security solutions and applications. • Explain common network communications technologies of IoT based on the solutions described in 2. • Explain key technologies and	• Text books, multi-media Projector, Lecture notes, Lab guide, simulation tools, microcontrollers, sensors Raspberry Pi, Arduino, etc	• Discover physical vulnerabilities in a mock-up IoT system with physical access to a Raspberry Pi and other tools. Perform a threat modelling exercise using simulation tool to model IoT physical vulnerabilities. • Perform a threat modelling exercise using simulation tool to model IoT application vulnerabilities.	• Demonstrate physical vulnerabilities in a mock-up IoT system with physical access to a Raspberry Pi and other tools. • Perform a threat modelling exercise using simulation tool to model IoT physical vulnerabilities.	Explain Sensors, Actuators and microcontrollers

	used NB-IoT communications technology in the IoT field	corresponding solutions of the widely used NB-IoT communications technology in the IoT field				
GENERAL OBJECTIVE: 3.0 Understand advanced communication technologies, IoT gateways' challenges in various industrial scenarios, and the capabilities and solutions offered by home gateways.						
8 – 9	3.1 Describe the popular communication technology 5G, including its development process, core technologies, application scenarios, and commercial solutions. 3.2 Describe difficulties and challenges faced by IoT gateways in industrial scenarios, edge computing, and mesh networking. 3.3 Describe capabilities of home gateways, as well as platforms and corresponding solutions.	• Explain the popular communication technology 5G, including its development process, core technologies, application scenarios, and commercial solutions. • Explain difficulties and challenges faced by IoT gateways in industrial scenarios, edge computing, and mesh networking. • Explain capabilities of home gateways, as well as platforms and corresponding solutions.	Text books, multi-media Projector, Lecture notes, Lab guide, simulation tools, microcontrollers, sensors Raspberry Pi, Arduino, etc	• Use Wi-Fi to implement a smart agriculture case, which involves collecting real-time data, responding to command delivery, and implementing device-cloud synergy. • Implement some other experiments like smart smoke detector, smart logistics, and smart street lamp	• Demonstrate on how to use Wi-Fi to implement a smart agriculture case, which involves collecting real-time data, responding to command delivery, and implementing device-cloud synergy.	

GENERAL OBJECTIVE 4.0: Understand IoT platforms, their positioning, capabilities, development process, and the role of operating systems						
10 – 11	4.1 Describe the IoT platforms, their positioning and capabilities	Explain Describe the IoT platforms, their positioning and capabilities	Text books, multi-media Projector, Lecture notes, Lab guide, simulation tools, microcontroller, Python IDE Raspberry Pi, Arduino, etc	Use Python to create a data pipeline to acquire, manipulate and visualize sensor-generated data	Guide the students to use Python to create a data pipeline to acquire, manipulate and visualize sensor-generated data	
	4.2 Describe the basic knowledge required for IoT platform development and the secondary development process.	Explain the basic knowledge required for IoT platform development and the secondary development process				
	4.3 Describe the development history of IoT operating systems	Explain the development history of IoT operating systems				
	4.4 Describe the problems faced by the IoT operating systems and devices	Explain the problems faced by the IoT operating systems and devices				
GENERAL OBJECTIVE: 5.0 Understand essential concepts for device development, sensor technologies, the kernel framework and APIs of Lite OS, and AT commands						
12 – 14	5.1 Describe the features of single-chip microcomputer and sensor technologies required for device development, kernel framework, and APIs of LiteOS.	Explain the features of single-chip microcomputer and sensor technologies required for device development, kernel framework, and APIs of LiteOS.	Text books, multi-media Projector, Lecture notes, Lab guide, simulation tools, microcontroller (Raspberry Pi, BearPi), LiteOS Raspberry Pi,	- Use LiteOS Studio to develop IoT devices and use LiteOS to control the IoT development board. - Implement device-cloud	- Demonstrate how to use LiteOS Studio to develop IoT devices and use LiteOS to control the IoT development board. - Guide the	
	5.2 Describe AT commands used in communications	Explain AT commands				

	modules interconnection testing	used in communications modules interconnection testing	Arduino, etc	connection based on NB-IoT Module	students to implement device-cloud connection based on NB-IoT Module	
	5.3 Describe network interconnection process using communications modules	Describe network interconnection process using communications modules				

LIST OF MINIMUM EQUIPMENT REQUIRED IN NETWORKING AND CLOUD COMPUTING LABORATORY

S/N	ITEMS	QUANTITY
HARDWARE SECTION		
1.	Computer Systems (Core i5, 8GB RAM) All networked	40
2.	Internet default gateway	1
3.	Server	2
4.	Multimedia Projector	1
5.	Printer	1
6.	UPS	40
7.	Generator (At le5KVA) or Solar System	1
8.	Raspberry Pi 4 8GB	20
9.	Arduino mega	10
10.	Arduino uno	10
11.	BearPi	10
12.	Router	2
13.	Switch	2
14.	Oscilloscope	2
15.	Network toolbox	10
16.	Twisted pair cable (CAT 6 and 7)	2 Rolls
17.	RJ-45 connector	2 Packs
18.	Soldering Station	10
19.	Set of screw drivers	20
SOFTWARE SECTION		
1.	Operating Systems (Windows and Linux)	
2.	Hypervisor (VMware, VirtualBox, Hyper-V etc.)	
3.	Virtual Machine (Windows, Linux etc.)	
4.	Network Monitoring applications (ManageEngine, SolarWinds, Datadog	

	etc.)	1 each
5.	Network Diagram tools (Microsoft Visio, LanFlow, Intermapper, Creately, SmartDraw etc.)	
6.	Cloud Platform (AWS/Azure)	
7.	Command-line tools (ping, traceroute/tracert, nslookup, Nmap etc.)	
8.	Wireshark	
9.	Port Scanners	
10.	Protocol Analyzer	
11.	Wi-Fi Analyzer	
12.	Bandwidth Speed Tester	
13.	PuTTY	
14.	Python	
15.	Simulation software (Packet Tracer, eNSP, GNS3)	
16.	Network mapper	
17.	SPSS	

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